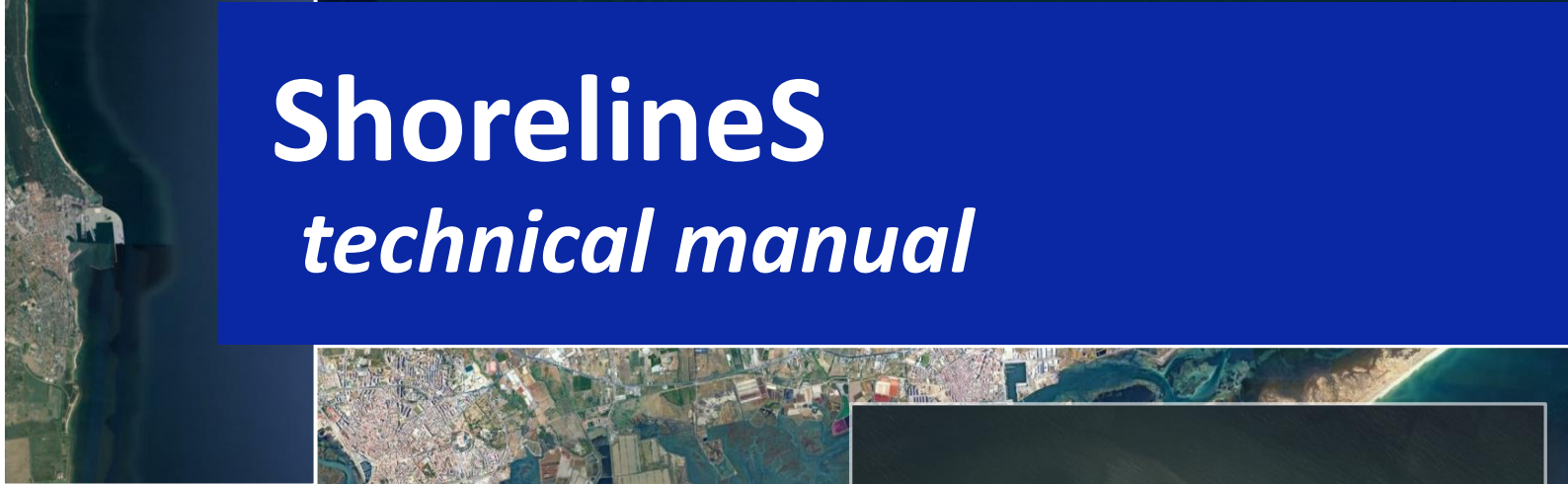
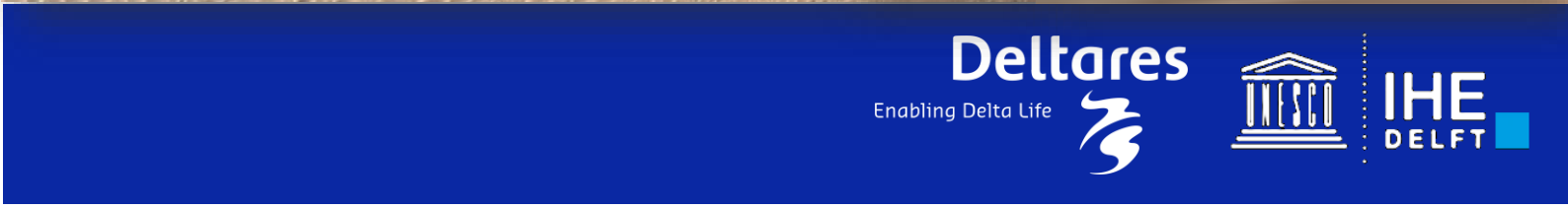


ShorelineS

technical manual

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ShorelineS : Technical manual

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Contents

1	Readers Guide	5
1.1	Scope	5
1.2	Readers guide	5
1.3	Technical reference version	5
2	Introduction to ShorelineS	6
2.1	The model in a nutshell	6
2.2	Typical applications	6
2.3	Capabilities	7
2.4	Required data	8
2.5	Main definition file	9
3	Spatial and temporal discretization	10
3.1	Coastline definition	10
3.1.1	Coastlines	10
3.1.2	Coastline evolution	11
3.1.3	Active profile height	12
3.1.4	Model grid generation	12
3.1.5	Spatially varying grids	13
3.1.6	Grid smoothing	13
3.1.7	Foreshore orientation	13
3.1.8	Cross-shore profile shape	14
3.2	Time frame	14
3.2.1	Specifying model duration	14
3.2.2	Adaptive coastal time step	14
3.2.3	Fixed coastal time step & aggregating conditions	15
3.2.4	Dune evolution time step	15
4	Wave, wind and water-level conditions	16
4.1	Waves	16
4.1.1	Types of wave input	16
4.1.2	Spatially varying conditions	17
4.1.3	Wave transformation for bulk transport formulations	17
4.1.4	Wave transformation and wave-driven current in profile mode	19
4.2	Tide	20
4.2.1	Required user input	20
4.2.2	Analytical tide model	21
4.2.3	Combined current velocity	22
4.3	Water-levels and surge	23
4.4	Wind	23
4.5	Climate change	23

5	Transport processes	25
5.1	Transport formulations	25
5.2	Lateral boundary conditions	28
5.3	High-angle corrections	29
5.4	Shadowing	30
6	Coastal interventions	31
6.1	Coastal structures	31
6.1.1	Defining groynes, breakwaters and revetments	33
6.1.2	Wave diffraction	33
6.1.3	Wave transmission	37
6.1.4	Bypassing at groynes	41
6.1.5	Revetment bypassing	42
6.1.6	Bypassing at hard layers	43
6.2	Nourishments	44
6.2.1	Beach nourishments	44
6.2.2	Shoreface nourishments	45
7	Cross-shore interactions	48
7.1	Barrier or spit overwash	48
7.2	Dune interaction	48
7.3	Cohesive / till dunes	53
8	Interaction between coast sections	54
8.1	Merging and splitting	54
8.2	Channel migration	55
9	Post-processing	56
9.1	Output data	56
9.2	Visualization of coastline and wave conditions	57
	References	58
A	Code structure	60
B	Overview of model input parameters	61

1 Readers Guide

1.1 Scope

This is a technical reference guide which provides a description of the input that is required for the ShorelineS model as well as the physics behind the many functionalities of the model, the different model settings and the numerical implementation.

1.2 Readers guide

To make this guide more accessible we briefly describe the contents of each chapter and the appendix.

Chapter 1: Provides an introduction to the manual

Chapter 2: Introduces the concept of ShorelineS and its basic use cases, as well as general information on required data and the main input definition method, which is relevant before reading the more in-depth chapters.

Chapter 3: Describes the spatial and temporal discretization. The grid definition and model generation are described, as well as choices for the active height, the foreshore orientation and smoothing of the grid. The way in which the coastline evolves over time is then discussed alongside the relevant input keywords for the model timestep and dune evolution timestep.

Chapter 4: The boundary condition definitions in the model are provided here for the waves, tides, water-levels, wind and climate change. Inputs can vary from single values to spatially varying time-series. Also, the process of refraction towards the depth-of-closure and point-of-breaking and alongshore interpolations are described.

Chapter 5: The available transport formulae are described. Furthermore, the settings for the lateral boundary conditions are provided. Corrections to the computed transport rates are also described, which can be relevant for regions with high-angle incident waves at spits and/or shadowing effects as a result of islands and structures.

Chapter 6: Coastal interventions such as revetments, offshore breakwaters and groynes can be added to the model, as well as nourishments. The schematization of these structures as well as their impacts on wave diffraction, sediment bypassing and local erosion resistance are described in this chapter.

Chapter 7: Cross-shore transport as a result of barrier overwash, dune growth and erosion and or cohesive dune elements is discussed here.

Chapter 8: This section focuses on the interaction between elements, such as the splitting and merging options, which allows elements to interact. The channel migration option is then described which allows tidal channels and river mouths to remain open.

Chapter 9: Relevant output data and the plotting/processing is discussed here.

1.3 Technical reference version

This manual applies to ShorelineS version 1.0 also known as the 'TKI' release.

2 Introduction to ShorelineS

2.1 The model in a nutshell

ShorelineS is an open-source numerical model for coastal planform evolution, based on one-dimensional equations for alongshore sediment transport and mass conservation, on a free-form grid that may consist of several coastal sections represented as strings of coastline points. The coastline sections can evolve freely and may include undulations, islands, spits, salient and tombolos; the sections may influence each other through shadowing and may merge or split. A range of hard elements or coastal structures are supported, such as groynes, T-groynes, offshore breakwaters and revetments; processes simulated include bypassing and diffraction. A range of longshore (bulk) transport formulas is available and soft engineering measures such as beach- and nearshore nourishments can be simulated.

2.2 Typical applications

The large-scale 'Sandmotor' nourishment that was constructed in 2011 shows the ability of the model to reproduce both the coastal erosion as well as the growth of the spit which encloses a lagoon (Figure 2-1).

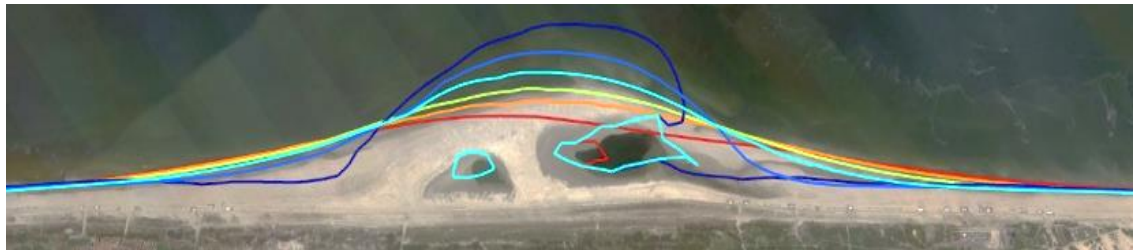


Figure 2-1 : Application of ShorelineS to model for the reshaping of the large-scale 'Sandmotor' nourishment in the Netherlands. The 2016 satellite image is shown.

The threat of coastal erosion is also very relevant to densely populated metropolitan areas. Such coasts have a history of coastal defence measures, with lots of structures and other protective measures already in place. It is therefore essential for the ShorelineS model to be able to deal with the complex interactions of the coastal structures with the sediment system of the coast. Groynes have been implemented in the code, which create sheltered areas with diffracted waves and bypass sediment under high wave conditions. An example of a complex coastline can be seen for Sitges (Figure 2-2).

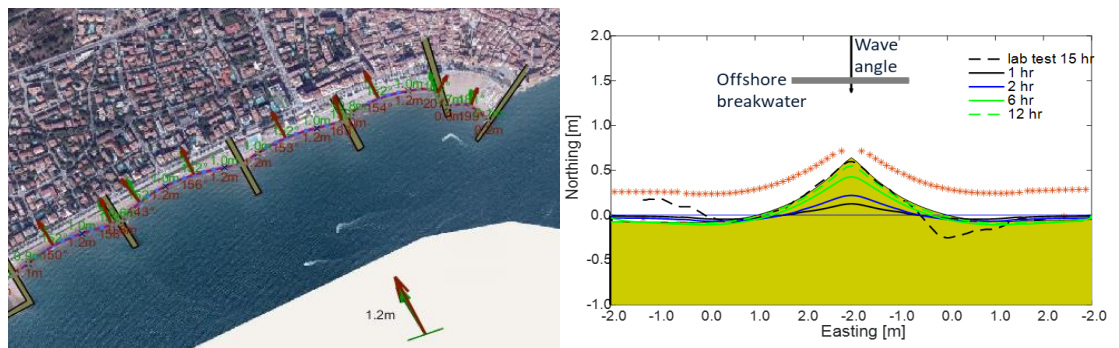


Figure 2-2 : Complex metropolitan coastline with structures at Sitges (left panel) and Creation of a tombolo behind an offshore breakwater (right panel)

Offshore breakwaters have been included in the model (Figure 2-2), which diffract waves from both ends and transmit also over and through the structures. A realistic tombolo shape can be created. Similarly, also the sheltering and diffraction of groynes is implemented in the model, as well as the obstruction and bypassing mechanisms at these structures. Even the effect of river outflow in an entrance of an estuary has been simulated for a case in Senegal, which shows that the model can easily be adapted to different situations. All these simulations can be prepared in hours to a few days and computations take minutes to a few hours.

Another example test case concerns the growth of a sandy spit in Angola with super-imposed groyne structures (Figure 2-3). Here the waves approach the coast under a very large angle from the southeast. Over time the spit has grown steadily over a period of many centuries. The difficulty for this case was not only making a good prediction of the northeast directed transport, but mainly of the growth direction of the spit. A very accurate wave transformation is needed to make a good prediction.



Figure 2-3 : Application of ShorelineS to model the growth of the Lobito spit.

2.3 Capabilities

The ShorelineS model implements unique functionality that is essential for reliable predictions of coastal changes due to natural processes, climate change or impacts of coastal protections.

1) Resolve sediment transport processes

The ShorelineS model computes (gradients in) wave- and tide-driven longshore transport along a 'grid' using multiple transport formulations (e.g. CERC, Kamphuis or Van Rijn, 2015). Equilibrium concepts are included for barrier overwash; dune growth and erosion are simulated using simple but realistic formulas for aeolian transport and wave impact.

2) Ability to handle complex coastlines

Barriers and elongated spit features can grow un-constrained in the model, as the numerical approach can handle high-angle wave coastline instabilities.

3) Interaction of multiple coastline segments

Segments of coastline can merge or split. Islands can merge, while lagoons can close or open.

4) Easy implementation of interventions

Various coastal protections can be handled (e.g. sand nourishments, revetments, groynes and breakwaters).

5) Efficiency for the user

The model is easy to apply. The grid is automatically re-generated every timestep, while the time-step is adjusted to the applied wave condition.

The model is unique when it comes to the ease of introducing arbitrarily shaped coastal sections, which will behave realistically. You see what you get. It is the link between the old-fashioned existing coastline models, which are fast but difficult to apply, and detailed but slowly evolving process-based models using a map-based grid. The ShorelineS model can be positioned as follows with respect to other models:

Table 1 : Inter-comparison of ShorelineS with regular coastline models and process-based field models

	Coastline models (e.g. Unibest)	ShorelineS	Field models (e.g. xbeach, Delft3D)
Processes	Beach transport (waves)	yes	yes, but slow
	Dune erosion/overwash	no	not or detailed ^{*2}
	Dune growth	no	experimental ^{*3}
	Spit & high-angle waves	yes	yes, takes long
	Climate change (decadal)	yes ^{*4}	yes, takes long
	Estuarine / tidal transport	inlet only ^{*5}	yes
	Sediment	sand & partially mud ^{*8}	sand & mud
Measure	Nourishments	multiple	multiple (hard after t0)
	Revetments	yes	yes
	Groynes	difficult ^{*6}	yes
	Offshore breakwaters	difficult ^{*6}	yes ^{*7}
Efficiency	Ease of use (complexity)	difficult	moderate / hard
	Time for model setup	weeks	>month
	Computational efficiency	quick	slow
	Adaptability of code	difficult	difficult
	Detail of the results	low	high

^{*1} : parameterized bypassing of sediment and dune erosion

^{*2} : only for some field models (e.g. XBeach, Delft3D FM)

^{*3} : experimental coupling (Delft3D FM-Aeolis)

^{*4} : for sandy coasts due to sea level rise or change in wave climate

^{*5} : river function to keep tidal inlets open

^{*6} : requires a model to compute the wave sheltering and diffraction

^{*7} : not easy to model the beach behind the structure

^{*8} : mud can be transported alongshore and retained at mud flat volume elements

^{*9} : both wave diffraction and transmission are accounted for

The code of the ShorelineS has been written with a clear open structure, in order to allow for the implementation of new developments. In this way the model code can be adapted to complex use cases.

2.4 Required data

Data required for the ShorelineS model would at least concern 1) measured coastlines, 2) wave climate conditions, 3) sediment parameters and 4) transport boundary conditions at the edge of the model. Depending on the application data may be needed of the interventions, such as 1) nourishments, 2) revetments, 3) groynes/port breakwaters and 4) offshore breakwaters. Also, the properties of 1) dunes, 2) spit geometry and 3) channel geometry may be needed in some cases. More details on the input needed for the model can be found in subsequent chapters.

ShorelineS has an extensive list of parameters that can be defined (see Appendix C). If they are not specified in the input file, then the default values will be used, which are also provided in the appendix. Any parameter specified as input for the model will replace the default values. Please take notice of any warnings the model provides when it initializes at the start of the model run, as it will provide a message if a keyword is added by the user that does not exist.

2.5 Main definition file

This chapter describes the input of the ShorelineS model. All these inputs are prescribed in the main function, called *ShorelineS*. This function is called with a state variable or structure *S*, which contains all necessary input for ShorelineS. In principle, there is no need to change anything in ShorelineS or lower functions, unless it is to fix bugs or add generic functionality.

The recommended way is to directly call “*runShorelineS(inputfile)*”. An example of the contents of *inputfile* would look like this for the circular island case:

```
%% MODEL INPUT PARAMETERS
reftime='2005-01-01';           % Reference time (i.e. 'yyyy-mm-dd')
endofsimulation='2009-01-01';
LDBcoastline='circle.txt';
WVCfile='wavedata.wvt';
d=6;                           % active profile height [m]
ds0=200;                       % initial space step [m]
..... more input
```

3 Spatial and temporal discretization

3.1 Coastline definition

3.1.1 Coastlines

The coastline in ShorelineS is described as a string of grid points (see Figure 3-1) that can move around, expand and shrink freely. The coastline points are assumed to be representative of the movement of the active coastal profile, and hence are situated at the MSL contour. The model can have multiple sections which may be closed (islands, lagoons). Sections can develop spits and other features and they may break up or merge as the simulation continues.

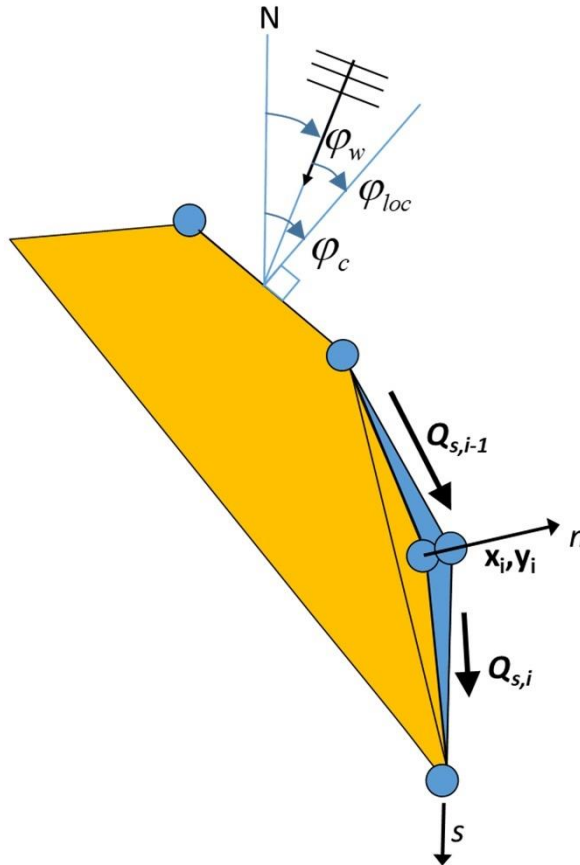


Figure 3-1 : Coastline-following coordinate system and definition of wave and coast angles. φ_c is the orientation of the shore normal with respect to North; φ_w is the angle of incidence of the waves with respect to North and φ_{loc} the local angle between waves and coast.

The coastline positions are given in two column vectors x_{mc} and y_{mc} , where the different coast sections are separated by NaN's. The directional convention is discussed below. If a section ends at the same coordinates as where it starts, it is treated as a cyclic section and may represent either an island or a closed lagoon. The coordinates may be in any Cartesian (metric) system. Structures are defined in a similar way, as two column vectors where different structures may be defined, separated by NaN's.

The coastline needs to be specified either as x-y vectors (using both parameters x_{mc} and y_{mc} which are $[1 \times N]$ row vectors) or in a text-file with the x and y-coordinates as 2 columns (in *LDBcoastline* with $[N \times 2]$ vector). Separate coastline segments can be specified, which requires

the coordinates to be separated by NaN's. Otherwise said, in all coastline sections, the land should be on the right when following the coastline points (default option). In many GIS systems the sea should be on the right; this is allowed if *gisconvention=1* is specified as a keyword. This orientation is internally converted to the ShorelineS convention. Only cartesian coordinate systems can be used, with units in meters.

Preferably only a single 'open' coastal element (or line-element) should be used, which should be the first element specified. Other elements should be closed (i.e. islands with begin and end point matching). The first element may also be a closed-element.

Land and lake segments can be defined. Assuming that *gisconvention=0*, the land-segments are defined clockwise, so lakes should be defined anticlockwise. The lakes should be inside coastal elements (so within an island or landward of an open-coast). The first element should always be a land-segment. So, lakes can only be the second element or later.

Note that the prescribed elements are regridded in the model, which may increase or decrease the resolution depending on the specified grid cell size (*ds0*). In general, it is preferable to have a rather smooth initial coastline. This aspect is especially important if it is a coast subject to high-angle wave incidence.

3.1.2 Coastline evolution

The basic equation for the updating of the coastline position is based on the conservation of sediment:

$$\frac{\partial n}{\partial t} = -\frac{1}{D_c} \frac{\partial Q_s}{\partial s} - \frac{RSLR}{\tan \beta} + \frac{1}{D_c} \sum q_i \quad (1)$$

where n is the cross-shore coordinate, s the longshore coordinate, t is time, D_c is the active profile height (m), Q_s is the longshore transport (m^3/yr), $\tan \beta$ is the average profile slope between the dune or barrier crest and the depth of closure, $RSLR$ is the relative sea-level rise (m/yr) and q_i is the source/sink term ($\text{m}^3/\text{m}/\text{yr}$) due to cross-shore transport, overwashing, nourishments, sand mining and exchanges with rivers and tidal inlets.

At each coastline point the local direction of the coast is determined as the normal to the reference line formed by the two adjacent coastline points (as a reference line). The longshore transport is then calculated for each segment. The transport gradient leads to the coastline point to move sea- or landward. The mass conservation equation is solved using a staggered forward time–central space explicit scheme (see Figure 1):

$$\Delta n_i^j = -\frac{1}{D_c} \frac{2(Q_{s,i}^j - Q_{s,i-1}^j)}{L_i} \Delta t \quad (2)$$

where j is the time step index, Δt is the time step (yr), i is the point/node index and L_i is the length of the considered grid element computed from $L_i = \sqrt{(x_{i+1} - x_{i-1})^2 + (y_{i+1} - y_{i-1})^2}$ and x_i and y_i are the Cartesian coordinates of point i . The stability criterion for the time step is discussed in Section 3.2. From the normal displacement it follows that the change in position of point i then becomes:

$$\begin{aligned}
\Delta x_i^j &= -\Delta n_i^j (y_{i+1} - y_{i-1}) / L_i \\
\Delta y_i^j &= \Delta n_i^j (x_{i+1} - x_{i-1}) / L_i \\
x_i^{j+1} &= x_i^j + \Delta x_i^j \\
y_i^{j+1} &= y_i^j + \Delta y_i^j
\end{aligned} \tag{3}$$

The scheme can be shown to be conserving the land area (Roelvink et al., 2020).

3.1.3 Active profile height

The active height (D_c) is used to describe the part of the profile that actively moves with the coastline. It is used in the code to translate the accumulation (or erosion) of sediment in m^3/yr in a grid cell (with length ds) to a coastline change in m/yr . It is very relevant to understand that there is a linear relationship between the active height and the response of the coast to gradients in alongshore sediment transport.

Typically, the active height is defined from the inner depth-of-closure (Hallermeier, 1981) to the dune toe. In some cases a wider area of the coastal profile is active, which may be the case for finer sediments and when rather long timescales (decades) are considered. In those cases either the outer depth-of-closure (Hallermeier, 1983) or mobility parameter related depth-of-closure (Hallermeier, 1978) can provide a better proxy. The active profile also includes the dune height in cases where the coast is very erosive, which means that dune erosion provides sediment to the wave-driven longshore currents.

The parameter d is used in the input file of ShorelineS for the definition of the active height. A spatially varying active height can be specified instead of a fixed value using $d=[x1,y1,hactive1;x2,y2,hactive2; \text{etc}]$.

3.1.4 Model grid generation

At start of the simulation the model will generate an equally spaced grid based on the specified grid cell size ($ds0$). The initial positions of the coastline are then interpolated along s to obtain the x and y positions of the grid points.

A typical grid cell size ($ds0$) is in the order of 50 to 100 m. Larger and smaller grid cell sizes can be used. Coarse grids are suited for large-scale problems where not too much detail is needed. Fine grid cells need to be combined with a small timestep and precise wave conditions. It is known that grid resolutions of $\leq O(10)$ meter pose some difficulties for a stable model run. Note that the grid cell size is not enforced strictly, but the model aims for generating approximately the specified grid cell size.

The model will automatically adapt the grid throughout the simulation, such that grid cell sizes will not be less than 70% or larger than 140% of the specified grid cells size ($ds0$). In cases where the grid sizes expand (e.g., at the tip of an expanding spit), new grid points are inserted where the grid size exceeds twice the initial prescribed grid size. Where the grid distances shrink (e.g., at an infilling bay or a shrinking spit) grid points are removed when the grid distance becomes less than 70% the original grid size.

Two 'grid regeneration methods' are available. By default the whole grid is re-interpolated when a cell is too small or too large (*griddingmethod*=2). A second method has been introduced which only splits or merges these cells that are actually too large or too small (and not redefine all cells), which can be activated using *griddingmethod*=1. This method is less diffusive.

3.1.5 Spatially varying grids

A spatially varying grid size can be used with gradual transitions if *griddingmethod=1* is used. This requires the *ds0* to contain a table with preferred grid-sizes at predefined locations, which will be interpolated smoothly over the model domain. For example: *ds0=[x1,y1,ds1; x2,y2,ds2; ... ; xn,yn,dsn]*.

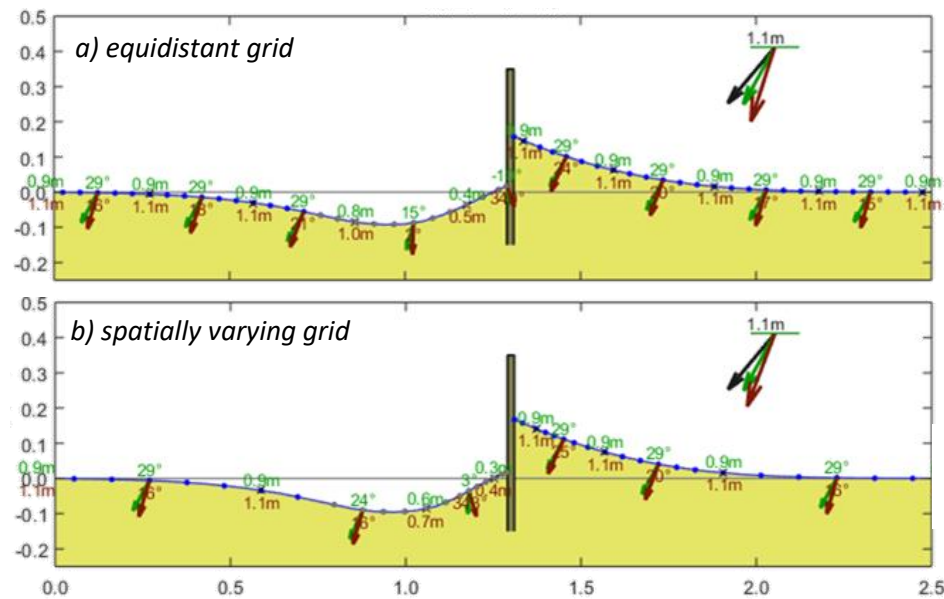


Figure 3-2 : Application of regular grid or a spatially varying grid for the same model with a groyne.

3.1.6 Grid smoothing

To avoid strong variations in grid size after inserting or extracting grid cells in expanding or shrinking sections, some smoothing of the s-grid can be applied when using *griddingmethod=2*. The smoothing factor has to be chosen carefully as too much smoothing may lead to a loss of planform area. The smoothing is used in 'active cells' only. The smoothing formulation applied in ShorelineS is a simple 3-point smoothing according to:

$$s_{i,smooth} = fs_{i-1} + (1-2f)s_i + fs_{i+1} \quad (4)$$

where *f* is a smoothing factor and *s* is the spatial position. By default the smoothing factor is zero to avoid artificial diffusion in the numerical scheme.

3.1.7 Foreshore orientation

Many coasts have a foreshore area outside the inner depth-of-closure that does not 1-on-1 respond to (small) changes in the coastline. Typically, the foreshore region is 1) beyond the depth-of-closure or 2) at a too large cross-shore distance from the shore to be meaningful (for very mild profiles). In any case, the foreshore region is expected to not respond too much to coastal change on the timeframe that is considered in the simulation (i.e. in years to decades). Such a foreshore region can be characterized with a rather static 'foreshore orientation', which is used in the model for the wave transformation from deep water (at the offshore point) to the nearshore point at the depth-of-closure (with *tdp* suffix), as discussed in Section 4.1. The foreshore orientation of the coastline can be specified as follows using the parameter *phif*:

- By default the foreshore orientation is not specified (*phif=[]*) which means that the initial coastline is assumed to be representative for the deeper foreshore, but it will not change over the simulation period.
- A fixed value can be prescribed for the foreshore orientation (e.g. *phif=270*; in °N describes a foreshore with a sea in the West and land in the East).
- A spatially varying foreshore orientation can be specified with prescribed foreshore orientations for locations along the coast, which is then interpolated over the model domain (using *phif=[x1,y1,phif1; x2,y2,phif2; etc]*).
- The current coastline orientation of the coast can also be used with some smoothing, which requires a cell-string as input (e.g. *phif={'gaussian',7}* which smooths over 7 cells).

3.1.8 Cross-shore profile shape

A cross-shore profile shape needs to be specified only when the tidal module is used. This needs to be done as a textfile in the keyword *profile*, with x and z coordinates as columns with positive values landward/upward. No header should be added. Furthermore, the keyword *dx* can be used to provide the grid cell size of the cross-shore profile.

3.2 Time frame

3.2.1 Specifying model duration

The start and end time of the model determine the model duration. The start time has to be entered as yyyy-mm-dd (e.g. *reftime='2020-01-01'*). The end of the simulation is added as yyyy-mm-dd (e.g. *endofsimulation='2040-01-01'*).

3.2.2 Adaptive coastal time step

A variable timestep is used by default, which is when parameter *tc* is not zero. The timestep is then automatically computed based on the transport rate. Since an explicit scheme is applied, the time step is limited by the following criterion (Vitousek & Barnard, 2015):

$$\frac{\varepsilon \Delta t}{\Delta s^2} < \frac{1}{2} \quad (5)$$

where the diffusivity ε is related to the maximum gradient of the sediment transport with respect to the wave angle relative to the coast, which can be approximated by:

$$\varepsilon_{\max} = 2Q_{\max} / D_c \quad (6)$$

where $Q_{\max}$ is the current maximum transport rate in the model. Therefore, the following is obtained:

$$\Delta t < \frac{D_c \Delta s^2}{4Q_{\max}} \quad (7)$$

This criterion can be restrictive for small grid sizes (e.g. less than 100m). Numerical stability is, however, guaranteed through this adaptive timestep.

It is possible to set the fraction of this automatic timestep with parameter *tc*. For example, a *tc=0.9* uses a timestep that is 90% of the maximum automatically computed timestep. Note that when a variable timestep is used the timestep will always be the minimum of the

automatically computed timestep and the time-intervals at which the wave conditions (in the input file) are specified. This ensures that wave conditions in the input files are never skipped.

3.2.3 Fixed coastal time step & aggregating conditions

For more control, a fixed timestep can be used by setting $tc=0$ and setting dt to a value in years (e.g. $dt=1/365$ to get a one day timestep). A typical timestep for a coastline model is 3 hours to 1 day. If you use a fixed timestep, then the model aggregates the input wave conditions within the fixed dt and applies this averaged wave condition. So, for a $dt=1/365$ (of one day) and a time-series that is defined every 3 hours, the model will take the average of all eight wave condition instances that occur within the considered time step of a day. It will add up the 'energy vectors' of all these wave conditions, take the mean, and translate the 'average energy vector' back to a weighted average condition.

3.2.4 Dune evolution time step

A dune sub timestep ($dtdune$) can be set for the aeolian and dune erosion computations which should be equal or smaller than the coastline time step. The $dtdune$ is defined as a fraction of a year. Thus, it allows for erosion processes at time scales smaller than the timescale at which coastline change is happening. If the $dtdune$ is not set, or left empty, then the minimum timestep of the timeseries of the waterlevel timeseries, wind timeseries and/or offshore wave timeseries is used. The timestep for the dune dynamics can never be larger than the timestep used for the coastline update, and is accordingly limited to a value smaller or equal to the coastline time step of the model.

4 Wave, wind and water-level conditions

4.1 Waves

4.1.1 Types of wave input

The wave climate at the offshore location can be specified in three ways with:

- 1) a static wave condition,
- 2) wave climates with probabilities per condition, and
- 3) time-series of wave conditions.

Static wave conditions

A fixed wave height, wave period and a range for the wave direction can be specified. At each time step a random wave direction will be chosen from this sector assuming a uniform probability distribution. A static wave height can be specified with the parameter *Hso* (significant wave height in meters), the wave direction with parameter *phiw0* (as wave direction in degrees following the nautical convention) and the wave period in parameter *tper* (as peak wave period in seconds). The spreading of the wave directions around the mean-wave direction can be adjusted with the parameter *spread*. The seeding number of randomized selection of wave climate conditions can be prescribed in *randomseed*, which allows for re-generation of the same random series. By default, this is off, meaning a different seeding number is taken for every new simulation.

Wave time-series

This is a sequential list of wave conditions, from which the model will interpolate the relevant conditions at each moment in time. A time-series can be specified using a file reference in the input parameter *WVCfile* with the extension '.wvt' (e.g. *WVCfile='filename.wvt'*). The time-series file has the format of four columns with date/time (as a string of *yyyymmddHHMM*), wave height (*Hs* in meters), wave period (*Tp* in seconds) and wave direction (*Dir* in nautical convention).

Wave climates

A wave climate consists of a number of wave conditions that are characterized by significant wave height, peak period, mean wave direction and a probability of occurrence. A wave condition is drawn randomly by the model at each time step taking into account the probability of the occurrence of the specific condition. Wave climates can be specified using a file reference (e.g. *WVCfile='filename.wvc'*). The model checks the file extension 'wvc' and then knows it is a wave climate. The wave climate file has the format of four columns with wave height (*Hs* in meters), wave period (*Tp* in seconds), wave direction (*Dir* in nautical convention) and probability (as a fraction of 1, or in percentage). The probabilities of the wave conditions are scaled with the sum of the probabilities of all the wave conditions in the specified file (assuming the sum is 100%). The seeding number for the randomized selection of wave climate conditions can be prescribed with the keyword *randomseed*, which allows for re-generation of the same random series. By default, the seed-number is not specified, meaning a different seeding number is taken for every new simulation. It is noted that a wave climate should not be combined with a variable timestep, but with a fixed timestep (i.e. with parameter *tc=0* and parameter *dt* at a timestep value), since a variable timestep would over-emphasize the low-energy conditions.

4.1.2 Spatially varying conditions

Spatially varying conditions can be applied for the wave climates and time-series. In general, using spatially varying nearshore wave conditions at the depth-of-closure is a good practice (i.e. close to the coast; possible even closer to the shore) which implicitly includes the transformation of offshore bathymetry in the wave boundary conditions, as this has been dealt with in the 2D wave computation. There are three options available within ShorelineS to specify the spatially varying wave conditions:

- Directly specifying references to multiple wave condition files with coordinates in the *WVCfile* parameter. This can be done using a cell-string as input for the *WVCfile*. With the first column being the names of the time-series/wave-climate files, and columns 2 and 3 respectively the x and y locations of the input wave files. As an example, it may look like *WVCfile*={*'example1.wvt',x1,x1; 'example2.wvt',x2,y2; etc*}. Similarly, the spatially varying wave climate can be specified.
- By directing the *WVCfile* to a file containing references to multiple WVT/WVC-files and their corresponding x and y locations. With the first column in this 'reference WVT/WVC-file' being the names of the time-series/wave-climate files, and columns 2 and 3 respectively the x and y locations of these input wave file As an example, it may look like *WVCfile*=*'referencing.wvt'*; And the content of the *'referencing.wvt'* contains *'example1.wvt',x1,x1* and *'example2.wvt',x2,y2* at the next line etc.
- A NetCDF file can be used to provide the nearshore conditions, which is recognized from the '.nc' extension in the *WVCfile*. It needs to contain variables 'station_x', 'station_y' for the location. Time/Date needs to be stored in a 'time' variable, which is in seconds with respect to a reference date (e.g. unix-epoch at 1979-01-01 00:00). Wave properties need to be stored in fields 'point_hm0', 'point_tp' and 'point_wavdir'.

The initial distance to each of the offshore/nearshore wave stations is stored, and later on used as a maximum distance for any object that may cast a shadowing of the waves at that coastal area.

It is noted that the selection of wave conditions for the boundary points has been synchronized, such that the same condition number is chosen for each location. For this reason, the applied wave climate conditions should be derived on the basis of the classification of one reference location. The seeding number of the randomized selection of wave climate conditions can be prescribed in *randomseed*, which allows for re-generation of the same random series. By default, this is off, meaning a different seeding number is taken for every new simulation.

After reading the spatially varying wave conditions at the specified stations, the model will be interpolated along the coastline at every timestep. For this purpose, two methods can be chosen with the keyword *interpolationmethod* which are:

- 'alongshoremapping' : this method projects the wave locations on the coast and then interpolates alongshore, which is the most accurate but also the slowest.
- 'weighteddistance' : with this approach a weighted average is taken of the nearest two wave locations based on the inverse of the distance to these points, which is less accurate but much quicker.

4.1.3 Wave transformation for bulk transport formulations

In case a bulk transport formulation is used, the prescribed wave conditions within ShorelineS will be transformed to the nearshore using Snell's law for the wave refraction, depth-induced breaking (based on a *gamma* breaking parameter) and shoaling based on the change in wave-celerity and group velocity at the considered depths. Both the wave height and direction are adjusted for the wave refraction. Wave heights may in landward direction reduce due to wave

refraction and breaking and increase due to shoaling. Note that the refraction from deep water to the toe of the dynamic profile is performed based on the assumption of parallel offshore depth contours. Wave conditions in the model are computed by the model at three cross-shore locations:

- the offshore or user-defined boundary condition depth, which is referred to with “o” symbol for each wave condition. The offshore depth should be specified by the user with the keyword *ddeep*. Note that *ddeep* is always the depth at the point at which the wave data were derived.
- waves in the nearshore location just outside the zone with the longshore drift, which are referred to with the “tdp” suffix for each condition. The nearshore depth should be specified by the user with the keyword *dnearshore*, which is typically at the depth-of-closure. The *ddeep* may be equal to *dnearshore* when the user defined waves were derived at the depth-of closure.
- a shallow-water location at the point of breaking “br” closest to the coast. This location is computed iteratively by the model.

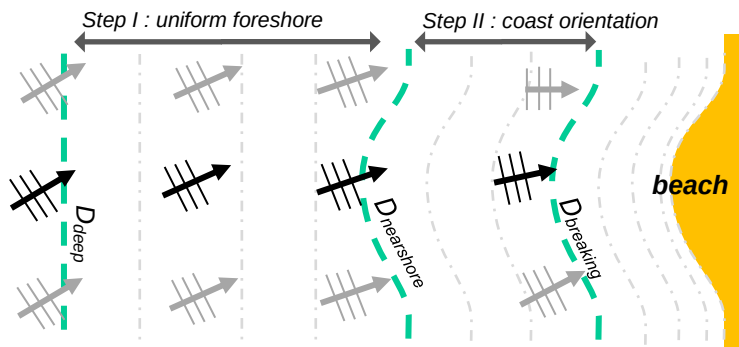


Figure 4-1 : Two-stage refraction procedure with first the transformation from the offshore location to the nearshore assuming a fixed foreshore, and a second step from the depth-of-closure to the point of breaking

The transformation is done in two steps from offshore to the nearshore, and then from the nearshore to the point of breaking. First the transformation over the ‘relatively static’ foreshore from offshore (o) to the depth-of-closure (tdp) is performed with Snell’s law, for which the wave angles (φ) are defined with respect to the foreshore orientation. The conditions at the depth of closure depend on the wave celerity (c), the wave group number (n) and the wave height (H_s).

$$\sin(\varphi_{tdp,o}) = c_{tdp} / c_o \sin(\varphi_o) \quad (8)$$

$$H_{s,cd} = \left(\frac{n_o c_o}{n_{tdp} c_{tdp}} \right)^{0.5} \left(\frac{\cos(\varphi_o)}{\cos(\varphi_{tdp,o})} \right)^{0.5} H_{s,o} \quad (9)$$

This is followed by the wave transformation from the nearshore point (tdp) to the point-of-breaking (br). Here φ is the wave incidence angle w.r.t. shore-normal of the coastline (instead of to the foreshore orientation).

$$\sin(\varphi_{br}) = c_{br} / c_{tdp} \sin(\varphi_{tdp,n}) \quad (10)$$

$$H_{s,br} = \left(\frac{n_{tdp} c_{tdp}}{n_{br} c_{br}} \right)^{0.5} \left(\frac{\cos(\varphi_{tdp,n})}{\cos(\varphi_{br})} \right)^{0.5} H_{s,tdp} \quad (11)$$

As an alternative to offshore conditions, it is possible to define nearshore conditions as input to the model, effectively skipping the first transformation step. For example, when waves from a 2D wave model are used at the depth-of-closure. In this case the offshore depth (d_{deep}) should have the depth of the output location where the waves were derived. The nearshore depth ($d_{nearshore}$) should be either the same or smaller than the offshore depth. It is noted that the basic bulk transport formulations CERC uses the offshore wave conditions that are specified (see Section 4.1).

4.1.4 Wave transformation and wave-driven current in profile mode

A wave transformation over the cross-shore profile is made when the tide module is used, with the Soulsby-Van Rijn formulation (keyword *trform* = *TIDEPROF*). In this case, the wave transformation is not based on just a refraction formulation, but the wave height variation is computed by solving the 1D wave energy balance:

$$\frac{\partial EC_g \cos \theta}{\partial x} = -D_w(E) \quad (12)$$

With E the wave energy, Cg the wave group celerity, θ the incident wave angle with respect to the shore-normal and D_w the dissipation of the waves due to breaking. The wave dissipation according to Baldock et al. (1998) can be written as follows:

$$\begin{aligned} D_w &= 0.25\alpha\rho g f_p \exp\left(-\frac{H_{\max}^2}{H_{rms}^2}\right) (H_{\max}^2 + H_{rms}^2) = \\ &= 2\alpha f_p \exp\left(-\frac{E_{\max}}{E_{rms}}\right) (E_{\max} + E_{rms}) \end{aligned} \quad (13)$$

With:

$$\begin{aligned} E_{\max} &= \frac{1}{8} \rho g H_{\max}^2 \\ H_{\max} &= \frac{0.88}{k} \tanh\left(\frac{\gamma k h}{0.88}\right) \end{aligned} \quad (14)$$

With α the wave breaking dissipation coefficient and f_p as a representative intrinsic frequency, g the acceleration of gravity [m/s²], ρ the density of water [kg/m³], H_{rms} the root mean square wave height [m], $H_{\max}$ the maximum wave height [m], γ the depth-induced wave-breaking parameter and k the wave number [m⁻¹]. A robust implicit, first order scheme for solving the cross-shore gradients in wave energy is then given by:

$$\frac{E_{i+1} Cg_{i+1} \cos \theta_{i+1} - E_i Cg_i \cos \theta_i}{\Delta x} = -2\alpha f_p \exp\left(-\frac{E_{\max}}{E_i}\right) (E_{\max} + E_{i+1}) \quad (15)$$

We can write the wave energy at the next grid point towards the coast as:

$$E_{i+1}^* = \frac{E_i Cg_i \cos \theta_i - \beta E_{\max}}{Cg_{i+1} \cos \theta_{i+1} + \beta} \quad (16)$$

With a dissipation factor β that is determined by the wave energy, wave breaking coefficient and the representative intrinsic frequency of the wave spectrum :

$$\beta = 2\alpha f_p \Delta x \exp\left(-\frac{E_{\max}}{E_i}\right) \quad (17)$$

For accuracy we then estimate the dissipation at the intermediate point as:

$$\beta^* = 2\alpha f_p \Delta x \exp\left(-\frac{2E_{\max}}{E_i + E_{i+1}^*}\right) \quad (18)$$

And the corrector step is:

$$E_{i+1}^* = \frac{E_i C g_i \cos \vartheta_i - \beta^* E_{\max}}{C g_{i+1} \cos \vartheta_{i+1} + \beta^*} \quad (19)$$

In order to obtain the wave-driven current we apply the bed shear stress by Feddersen et al. (1998) and equate this to the longshore wave pressure force F_y :

$$\frac{F_y}{\rho C_f} = U_{rms} \bar{v}_w \left(1.16^2 + (\bar{v}_w / U_{rms})^2\right)^{1/2} \quad (20)$$

With U_{rms} the orbital velocity of the waves and V_w the wave-driven current velocity [m/s].

We can rewrite this implicit relationship as a quadratic equation in $\bar{v}_w^2$ and solve it:

$$\bar{v}_w = \sqrt{\frac{-b + \sqrt{b^2 - 4ac}}{2a}}, \text{ with: } a = 1/U_{rms}^2, \quad b = 1.16^2, \quad c = -\left(\frac{F_y}{\rho C_f U_{rms}}\right)^2 \quad (21)$$

4.2 Tide

Tidal currents and water-levels can be added in the ShorelineS model. The tidal module is switched on when the transport formulation *TIDEPROF* is used (with the keyword *trform*). This allows the model to be used for the evaluation of many coastlines around the world, as many beaches are mixed tide and wave influenced. Three aspects are considered here:

- 1) reading tidal current data of user-defined stations,
- 2) an analytical tide model to reconstruct the tidal currents over the whole cross-shore profile.
- 3) computing the effect of combined tide and wave driven sediment transport. This is discussed in the section on the *TIDEPROF* transport formulation.

4.2.1 Required user input

As input the model requires a set of tidal constituents of the M2 and M4 tides at a limited number of representative alongshore locations at depths of about 5 meter. These tidal constituents need to be derived at the considered locations from the water-level time-series of a 2D tide-model that is driven by M2 and M4 tide, and specified in a table in a tide-input file (*tidefile*). The input is:

- x and y locations of the stations ('x_stat' and 'y_stat'), which are the first two columns
- vertical amplitude of M2 and M4 tide components ('eta' in meter) in 3rd and 4th column,

- longshore gradient of eta ('detads') in the 5th and 6th column,
- phase of vertical tide components ('phi' in degrees) in the 7th and 8th column,
- alongshore wave number of tidal components ('k' in radians/meter) in column 9 and 10.
- mean longshore surface slope ('surfslope') in the 11th column.

4.2.2 Analytical tide model

An analytical method has been developed to assess the cross-shore distribution of the tidal current. First the theoretical derivation of the alongshore and cross-shore tidal currents are discussed, after which the wave transformation over the profile and the combination of the velocities is explained.

Longshore currents

The tide-driven current is solved analytically over an arbitrary, longshore uniform, profile. The linearized equation for the longshore momentum balance is:

$$\frac{\partial v}{\partial t} = -g \frac{\partial \eta}{\partial s} - \frac{\tau_b}{\rho h} + \frac{F_s}{\rho h} \quad (22)$$

With v the tidal velocity [m/s], t is time [s], g the acceleration of gravity [m/s²], η the water-level difference [m], s the alongshore distance [m], τ_b the shear stress due to friction [N/m²], ρ the density of water [kg/m³], h the water depth [m] and F_s a pressure force acting on the water column [N/m²]. The shear stress is linearized:

$$\frac{\tau_b}{\rho h} = \frac{\rho C_d |v| v}{\rho h} \approx \frac{\lambda}{h} v \quad (23)$$

With C_d the drag coefficient and λ the friction parameter [m/s]. A periodic solution is used in the model, of this form:

$$\begin{aligned} \eta &= \hat{\eta} \cos(\omega t - ks - \varphi) + s_s s \\ v &= v_1 \cos(\omega t - ks - \varphi) + v_2 \sin(\omega t - ks - \varphi) + \bar{v} \end{aligned} \quad (24)$$

With ω the angular velocity [rad/s], Φ the phase of the wave [rad], k the wave number [rad/m], η the water level elevation with respect to the mean [m] and s_s the factor for the residual slope of the water surface. The gradients in y en t are given by:

$$\begin{aligned} \frac{\partial \eta}{\partial s} &= +k \hat{\eta} \sin(\omega t - ks - \varphi) + \frac{\partial \hat{\eta}}{\partial s} \cos(\omega t - ks - \varphi) + s_s \\ \frac{\partial v}{\partial t} &= -v_1 \omega \sin(\omega t - ks - \varphi) + v_2 \omega \cos(\omega t - ks - \varphi) \end{aligned} \quad (25)$$

The following periodic solution for the tidal velocities and phases was found based on the linearized equation for the longshore momentum balance and linearized shear stresses (λ) for an arbitrary, longshore uniform, profile.

$$\begin{aligned}
v_1 &= \frac{g}{\omega} \left(\frac{1}{1+\phi^2} k\hat{\eta} - \frac{\phi}{1+\phi^2} \frac{\partial \hat{\eta}}{\partial s} \right) \\
v_2 &= -\frac{g}{\omega} \left(\frac{\phi}{1+\phi^2} k\hat{\eta} + \frac{1}{1+\phi^2} \frac{\partial \hat{\eta}}{\partial s} \right) \\
\bar{v} &= -\frac{gh}{\lambda} s_s \\
\phi &= \frac{\lambda}{\omega h}
\end{aligned} \tag{26}$$

The value of the friction parameter λ is given by solving a quadratic equation in $\hat{v}^2$:

$$\begin{aligned}
\lambda &= 3/8 \cdot \pi C_d \hat{v} \\
\text{with: } \hat{v} &= \sqrt{\frac{-b + \sqrt{b^2 - 4ac}}{2a}}, \quad a = \left(\frac{3/8 \cdot \pi C_d}{\omega h} \right)^2, \quad b = 1, \quad c = -\left(\frac{g}{\omega} \sqrt{k^2 \hat{\eta}^2 + \frac{\partial \hat{\eta}^2}{\partial s}} \right)^2
\end{aligned} \tag{27}$$

In this way an analytical solution is obtained that only depends on the water depth, which varies with tidal phase and cross-shore distance. Adding higher components such as M4 is simply a matter of superposition of the signal for each tidal component, given the linear assumption. The computation of λ can be based on the total rms amplitude of all considered tidal components.

Also, the cross-shore velocities can be reconstructed when the gradient in the alongshore tidal velocity and the change in the tidal water-level is known.

$$\frac{\partial \eta}{\partial t} + \frac{\partial v}{\partial s} + \frac{\partial u}{\partial n} = 0 \Rightarrow \frac{\partial u}{\partial n} = -\frac{\partial \eta}{\partial t} - \frac{\partial v}{\partial s} \tag{28}$$

$$u_n = -\int_0^n \left(\frac{\partial \eta}{\partial t} + \frac{\partial v}{\partial s} \right) dn = -n \frac{\partial \eta}{\partial t} - \int_0^n \frac{\partial v}{\partial s} dn \tag{29}$$

Where u_n is the cross-shore-velocity [m/s] and $\partial u / \partial n$ the cross-shore gradient in the cross-shore velocity [1/s].

4.2.3 Combined current velocity

The combined velocity due to tide, tide residual and wave-driven current as a function of x and tidal phase is obtained by simple addition:

$$v(x, t) = v_{tide} + \bar{v}_w \tag{30}$$

An efficient procedure to solve both the tidal and wave-driven currents through the tide and over the profile is as follows. We start from a given bed level:

$$z_b = f(x) \tag{31}$$

And interpolate this to a uniform grid with grid size Δx . The water level variation throughout the tidal cycle is given by:

$$\eta(t) = \hat{\eta}_{M2} \cos(\omega_{M2}t - \varphi_{M2}) + \hat{\eta}_{M4} \cos(\omega_{M4}t - \varphi_{M4}) \quad (32)$$

Where t is given in a number of discrete steps per tidal period.

If we put $\eta(t)$ in a row vector and $z_b(x)$ in a column vector then the resulting waterdepth is:

$$h(x,t) = \eta(t) - z_b(x) \quad (33)$$

The tidal velocity can be solved using these formulations. For the wave-driven currents we can compute the dispersion relation, wave refraction and resulting wave-driven currents on the whole matrix, while we can solve the energy balance by stepping in x-direction, taking along all time points at once (Section 4.1.4).

The longshore transport for each x and t is computed given the matrices of h and v , plus the wave parameters, also given as matrices. We apply the Soulsby-van Rijn formula (see *TIDEPROF* formulation in Section 5.1):

4.3 Water-levels and surge

Surge levels (which need to include also the tide level) in the model are needed when the dune interaction module is applied. They are specified using a time-series of waterlevels in a text-file (*WATfile*) or as a fixed value (*WLO*). The water level time series file should have a .WAT extension and contain a two columns [Nx2] with date/time in the first column (in *yyyymmddHHMM*) and waterlevel as the second column (in meter relative to MSL). A spatially varying water level field can be used by referencing multiple files either directly in the keyword *WATfile* with {'file1', x1,y1; ...} or by using a file which contains these references (i.e. at every row the reference to a .WAT-file and the coordinates: file1 x1 y1 etc).

4.4 Wind

Dune growth is driven by wind forcing, which requires wind timeseries. This wind time-series file can be specified with the keyword *WNDfile* and is a text-file with .WND extension. The file contains data with a [Nx3] format, with in these columns the date/time in 'yyyymmddHHMM', wind velocity [in m/s] and direction [in degree North]. Alternatively, a static wind climate can be specified with the wind velocity (keyword *uz*), wind direction (keyword *phiwind0*). In all cases a height needs to be specified at which the wind is defined (keyword *z*). Furthermore, a wind drag coefficient needs to be specified (keyword *Cd*). A spatially varying wind level field can be used by referencing multiple files either directly in the keyword *WNDfile* with {'file1', x1,y1; ...} or by using a file which contains these references (i.e. at every row the reference to a .WND-file and the coordinates: file1, x1, y1 etc).

4.5 Climate change

Sea level rise can be a constant rate (e.g. *ccSLR*=0.002 m/yr) or a table is used in *ccSLR* with the sea level rise with respect to the initial SL value against time [Nx2]. Wherein 'time in 'yyyymmdd' format' and 'sea level with respect to initial situation' are specified (in meter). Note that the rates per year are then computed automatically. The 'slope angle', as specified in keyword *tanbeta*, is used for the Bruun coastal retreat rule.

A climate impacted increase of the wave height can be a constant rate per year (e.g. *ccHS*=0.001, which is +0.1% increase in HS per year). Or alternatively, a table is used in *ccHS*

with a time varying wave height change [Nx2]. Wherein the first column has the 'time in 'yyyymmdd' and the second column the 'relative change in wave height w.r.t. initial situation'.

The climate impacted change in wave direction can be a constant rate (e.g. $ccDIR=0.05\text{ }^{\circ}/\text{yr}$) or a table with the date/time and 'wave direction change with respect to the initial situation' [Nx2]. Wherein 'time in 'yyyymmdd' and 'relative change in wave direction w.r.t. initial situation' as a number of degrees.

5 Transport processes

The coastline changes within ShorelineS are driven by wave-driven currents, tidal currents and cross-shore processes due to wind or storms. The available transport formulations are discussed in Section 5.1. Section 5.2 provides an overview of the lateral boundary conditions that can be provided at the start and end of open coastal elements. High-angle corrections for high-angle incident wave conditions are discussed in Section 5.3. Coastal segments and structures can also block the wave energy at sections of coast, which is accounted for by a shadowing function (Section 5.4). Note that wave diffraction and sediment bypass processes are discussed in Chapter 6, while cross-shore transport is discussed in Chapter 7.

5.1 Transport formulations

A couple of transport formulations can be chosen. The transport formulations can be subdivided in formulations that use 1) directly the user-defined offshore wave condition, 2) the nearshore wave conditions at the point-of-breaking as computed using wave refraction and shoaling and 3) wave transformations and transports that are computed over the whole cross-shore profile.

The CERC formulations were developed by the US army Corps (USACE, 1984). Different versions of this formulation have been used over time, of which three common versions are included in ShorelineS. In addition, two very similar formulations are available, which include the sensitivity to more physical parameters, which are the Kamphuis (1991) and Mil-Homens (2013) formulations. The CERC1 and CERC2 use offshore wave conditions, while all other formulations (e.g. CERC3, KAMP and Mil-Homens) use transports at the point-of-breaking. However, in all cases the dependence of the sediment transport on the offshore wave direction is rather similar, with a maximum transport occurring at an offshore angle of 40° to 45° from shore-normal wave incidence. The transports of each formulation can be calibrated to match the local transport rates using the 'qscal' parameter.

It is noted that when diffraction is used, an additional term for the gradient in wave energy is resolved in the 'transport computation' which resembles the effect of residual water-level setup driven circulation.

Wave-driven longshore transport using offshore waves (CERC1 and CERC2)

CERC1 is the simplest formula and is mainly meant for illustrating the principles of the behavior of the coastline model. The transport rate Q_s [m³/yr] in CERC1 is computed as follows:

$$\text{CERC 1 : } Q_s = bq_{scal}H_{S,off}^{5/2} \sin 2(\phi_{off}) \quad (34)$$

With $H_{S,off}$ as the significant wave height at deep water [m], the ϕ_{off} as the offshore wave direction with respect to the shore-normal of the coastline [°] and b the CERC parameter, defining the magnitude of the transport [m^{0.5}/yr], which can be provided as a keyword. This parameter should include the conversion from seconds to years. The default value is 1 million for b , but this value has to be set by the user for every case. Calibration is possible using the q_{scal} parameter [-] (keyword 'qscal'), which is by default set at 1.

CERC2 is derived from the official CERC1 formula to include the effect of refraction and shoaling into the formula. Its behavior is quite similar to CERC1 though. The CERC2 allows for a direct comparison with the Coastal Evolution model which utilises it (Ashton and Murray,

2006). The sensitivity to the wave period (T_p in [s]) is also part of the CERC2 formulation, as it affects the wave transformation and subsequently the transport rates. The CERC2 transport formulation is parameterized as follows:

$$\text{CERC 2 : } Q_s = K_1 q_{scal} t_{sc} H_{S,off}^{2.4} T_p^{0.2} \cos^{1.2}(\phi_{off}) \sin(\phi_{off}) \quad (35)$$

with:

$$K_1 = \left(\frac{\sqrt{g\gamma}}{2\pi} \right)^{\frac{1}{5}} K_1,$$

$$K_2 = \frac{K_0}{16(1-p)(\rho_s - \rho_w) / \rho_w} \sqrt{\frac{g}{\gamma}}$$

This formulation computes the transport scaling parameters (K_1 and K_2) depending on the wave breaking parameter (γ). The ρ_w and ρ_s are the density of water and sediment [kg/m³], while p is the porosity. An input value for K_0 of 0.39 m^{0.5}/s is used for CERC2. Note that K_2 is the coefficient for the formulation with wave parameters at the point-of-breaking, and the coefficient K_1 is the corrected coefficient for situations when offshore wave parameters are used (i.e with implicit wave transformation). The sediment transport rate is scaled to yearly rates with t_{sc} (=365·24·60·60).

Wave-driven longshore transport using nearshore waves (CERC3, Kamp, Milh)

The CERC3 formula is widely used in models worldwide such as GENESIS. It is the more common version of the CERC transport formula (USACE, 1984) which is applied after the wave transformation has taken place (e.g. in a 2D wave model). It is still a very basic formulation, but then for the nearshore waves.

$$\text{CERC 3 : } Q_s = K_2 q_{scal} t_{sc} H_{S,br}^{5/2} \sin 2(\phi_{br}) \quad (36)$$

The refracted wave height at the point-of-breaking ($H_{S,br}$ in [m]) and the wave direction with respect to the shore-normal (ϕ_{br} in [°]) at this location determine the transport rate. The scaling parameter K_2 is applied in a similar way as in the CERC2 formulation, but then using a K_0 parameter of 0.35 m^{0.5}/s as defined in the Shore Protection Manual (USACE, 1984).

The Kamphuis formulation adds extra input parameters such as beach slope and grain size in order to provide more accurate results with less calibration effort (Kamphuis, 1991). And similarly, the Mil-Homens (2013) formulation uses these parameters but with re-calibrated parameters.

$$\text{KAMP : } Q_s = 2.33 q_{scal} t_{sc} K_3 H_{S,br}^2 T_p^{1.5} \tan_{\beta}^{0.75} D_{50}^{-0.25} \sin^{0.6}(2\phi_{br}) \quad (37)$$

$$\text{MILH : } Q_s = 0.15 q_{scal} t_{sc} K_3 H_{S,br}^{2.75} T_p^{0.89} \tan_{\beta}^{0.86} D_{50}^{-0.69} \sin^{0.5}(2\phi_{br}) \quad (38)$$

with:

$$K_3 = \frac{1}{(1-p)(\rho_s - \rho_w)}$$

With the D_{50} being the median grain diameter [m], $\tan_{\beta}$ is the mean bed slope (beach slope in the breaking zone).

Tide and wave-driven longshore transport using nearshore waves (VR14)

The Van Rijn (2015) formulation (*trform*='VR14') is a parameterization of the TRANSPOR2004 formulation. It computes the transport (Q_s [m³/yr]) in a very similar way as the Mil-Homens and

Kamphuis formulation, but requires also a swell wave parameter (K_{swell} [-]), which depends on the occurrence of the swell conditions (P_{swell} in percentage of the year). In addition to the computed alongshore current due to waves (v_{wave}) also the tide-driven flow velocity can be specified (v_{curr}), which is a capability that will be added in the future.

$$VR14 : \quad Q_s = 0.0006 \cdot q_{scal} t_{sc} K_4 K_{swell} \rho_s \tan \beta^{0.4} D_{50}^{-0.6} H_{s,br}^{2.6} v_{total} \quad (39)$$

with:

$$v_{total} = v_{wave} + v_{curr}$$

$$V_{wave} = 0.3 (g H_{s,br})^{0.5} \sin(2\phi_{br})$$

$$K_{swell} = 0.015 P_{swell} (1 - 0.01 \cdot P_{swell})$$

$$K_4 = 1/(\rho_s (1 - p))$$

Tide and wave-driven longshore transport over a cross-shore profile (TIDEprof)

The wave transformation and transports are computed over the whole cross-shore profile for the Soulsby-VanRijn formulation with tide ($tform='TIDEPROF'$). This requires not only the specification of the waves at the boundary, the bed-friction coefficient (C_f), a minimum water depth (h_{min}) and closure depth for the tidal transport ($h_{closure}$). The depth-of-closure indicates till what offshore extent the tide-induced bed changes are still of relevance for the coastline. Also, a cross-shore profile needs to be specified as a textfile in the keyword *profile* (with x and z coordinates as columns with positive values landward/upward). The keyword *dx* can be used to specify the cross-shore grid cell size (see Section 3.1.8).

The formulation of Soulsby van Rijn (1997) is solved for each cross-shore grid cell of the coastal profile, which combines a bed load ($Q_{s,bed}$) and suspended load transport ($Q_{s,sus}$) component:

$$Tideprof : \quad Q_s = Q_{s,bed} + Q_{s,sus} \quad (40)$$

with:

$$Q_{s,bed} = 0.005 A_{cal} t_{sc} h \left(\frac{D_{50}/h}{\Delta g D_{50}} \right)^{1.2} u \cdot \xi \quad (41)$$

$$Q_{s,sus} = 0.012 A_{cal} t_{sc} D_{50} \frac{D_*^{-0.6}}{(\Delta g D_{50})^{1.2}} u \cdot \xi \quad (42)$$

With D^* being the dimensionless grain size, D_{50} the median grain diameter [m], u the velocity of the tide [m/s], h the water depth [m], A_{cal} a calibration coefficient (default is 0.2) and ξ the mobility of the sediment. The Δ is the ratio of the immersed sediment density over the water density $\Delta = (\rho_s - \rho_w)/\rho_w$.

$$D_* = \left[\frac{g \Delta}{\nu^2} \right]^{1/3} D_{50} \quad (43)$$

$$u_{rms} = \frac{\pi H_{rms}}{\sqrt{2} T_p \sinh(kh)} \quad (44)$$

$$\xi = \left(\sqrt{u^2 + \frac{0.018}{C_F} u_{rms}^2} - u_{cr} \right)^{2.4} \quad (45)$$

The mobility parameter (ξ) depends on the velocity of the tide (u), the wave height (H_{rms}), wave orbital velocity (u_{rms}), the friction coefficient (C_F) and the critical velocity for mobilizing sediment (u_{cr}). The critical velocity is computed on the basis of the median and 90th-percentile of the sediment grain size (D_{50} and D_{90}) and the water depth (h). The Von Karman constant (κ), water depth (h) and the roughness height (z_0) are important for the friction factor.

$$u_{cr} = \begin{cases} 0.19D_{50}^{0.1} \log_{10}(4h / D_{90}) & \text{if } D_{50} \leq 0.5 \text{ mm} \\ 8.5D_{50}^{0.6} \log_{10}(4h / D_{90}) & \text{if } 0.5 < D_{50} \leq 2 \text{ mm} \end{cases} \quad (46)$$

$$C_F = \left[\frac{\kappa}{\ln(h / z_0) - 1} \right]^2 \quad (47)$$

Besides the wave conditions at the seaward side of the profile, also, a set of tidal constituents of the M2 and M4 tides needs to be provided at a number of alongshore locations at depths of about 5 meter. See Section 4.2 for more information on the tide input.

5.2 Lateral boundary conditions

Various lateral boundary conditions were implemented in the model to represent a variety of coastal situations. Boundaries can be specified at the start and end of all non-cyclic/open coastal elements using the keywords *boundaryconditionstart* and *boundaryconditionend*. Cyclical coastal elements are detected by the model, as it checks whether a section end point is near the section start point. The options for the non-cyclical boundaries are :

- **closed boundary**
This type closes the boundary for sediment transport (when 'closed') or predefines a transport that is enforced at this boundary (which can be done using for example {'closed',25000} for a 25 thousand m³/yr transport).
- **fixed coastline position**
A stationary position of the coastline is obtained with a 'fixed' or 'Neumann' boundary, which effectively ensures that the transport at the boundary is the same as the transport in the adjacent transport point of the coastline, resulting in a locally zero transport gradient.
- **fixed coastline orientation**
In this case the orientation is the initial orientation throughout the simulation (using 'angleconstant') or a predefined orientation that is specified by the user (e.g. {'angleconstant',310} for a fixed 310°N angle at the boundary).
- **periodic boundary condition**
the start and end boundary of the model can also be connected using a 'periodic' boundary. The computed transport rates at the start and end grid points will be averaged throughout the simulation.

The model stores the initial coastline orientation of the boundary points. Grid cells at the boundary will then move perpendicular only to this initial orientation of the coastline.

A number of tests were performed with combinations of the implemented boundary conditions for a static uniform wave climate (Figure 5-1), which show they work well.

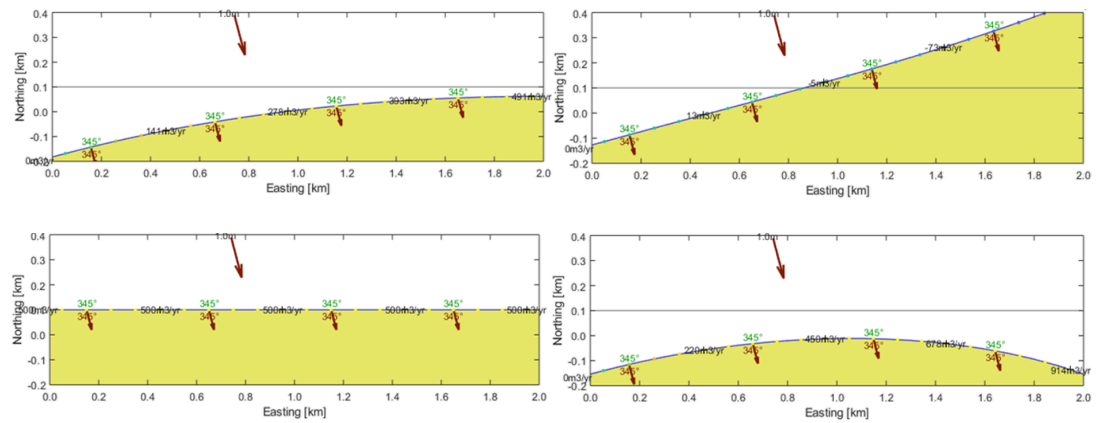


Figure 5-1 : Application of lateral boundary conditions at the model for a straight coast under a uniform wave condition of 1 meter from a mildly oblique angle, with the following combinations of boundary conditions: Top-left panel : Closed – Angleconstant; Top-right : Closed – Close 200,000 m³/yr, Lower-left : Angleconstant - Fixed; Lower-right : Closed – Angleconstant -20°.

5.3 High-angle corrections

High-angle instabilities may develop when waves from very oblique incidence angles ($\sim 40^\circ$ or more) approach the coast (Ashton et al., 2001). In those cases, small perturbations tend to grow, because there is too little wave energy at the downdrift side of a coastal perturbation to transport the sediment away. This is not just a numerical feature of models, but also a physical phenomenon that can cause spit features to grow or induce shoreline undulations.

A special treatment takes care of these so-called high-angle instabilities, which allows spits to develop. The model caps these instabilities by recognizing the locations with transitions from low to high-angle wave incidence. In these cases where the local angle exceeds the critical angle, the transport at the downdrift point is set to the maximum transport. In this way spits may grow, as can be seen for the Sand Motor and natural flying spits in Namibia, but the model remains stable. Figure 5-2 illustrates the effect of this treatment, where a central scheme would lead to unstable behaviour, the local upwind treatment ensures a smooth development into a spit. The physics in the model are similar to those in Ashton et al. (2001, 2016), and Ashton and Murray (2006), and therefore it inherits most of the behaviour of the Coastal Evolution Model (CEM). Within ShorelineS this behaviour is achieved with a vector-based rather than a grid-based approach, which is more efficient, especially when large areas need to be covered.

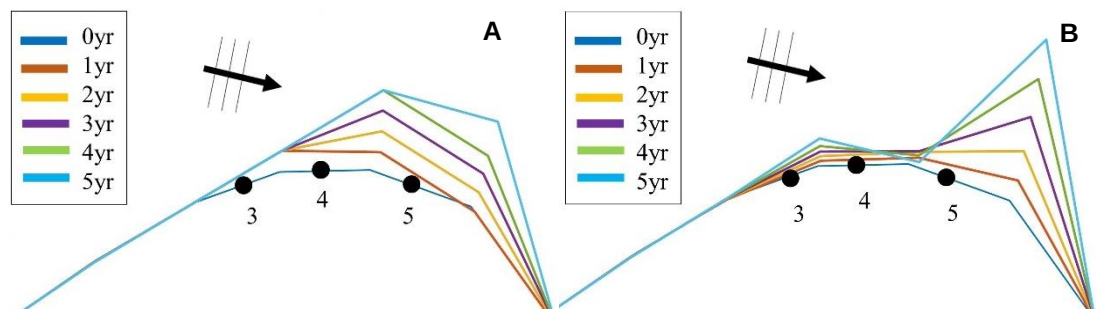


Figure 5-2. Example of high-angle instability with standard central scheme (A) and upwind scheme (B).

It is possible to spread the excess sediment at the point where low-angle waves become high-angle over more than a single point by specifying the keyword *twopoints*. It is advised to use

twopoints=1 to spread the sediment at high-angle transitions also over the second downdrift cell, which provides a more smooth coastline response. In addition, the *maxangle* can be adjusted to limit the angle change between grid cells. By default *maxangle=60°*.

With *plotUPW=1* it is possible to plot the locations where the high-angle correction is applied, which show up as green and red squares.

5.4 Shadowing

Coastal elements can block wave energy from reaching other coastal sections, which is very relevant for the behaviour of coastlines. As wave conditions can continuously change, also the shadowing of coastal elements needs to be re-evaluated every time step. Shadows are computed as a result of coastal elements/island and coastal structures, both for the coastline points as well as the transport points of the grid. Sediment transport is set to zero at these locations, unless they are also influenced by diffraction.

6 Coastal interventions

6.1 Coastal structures

Structures can be added in the model to provide wave shielding, block or reduce alongshore transport rates and to prevent erosion. The following types of structures are available:

- **Offshore breakwaters**

These are hard coastal structures that are constructed coast-parallel in shallow to intermediate water depths up to the depth-of-closure. Breakwaters have diffraction points at both sides, which affect the waves in the sheltered area (Section 6.1.2). Furthermore, they can transmit part of the wave energy over the structure (Section 6.1.3). Offshore breakwaters should be located offshore at a somewhat similar distance from the coastline, and not be shading each other. The offshore breakwater preferably does not cross the coastline. If it crosses the coastline twice then the model will assume it is a groyne instead.

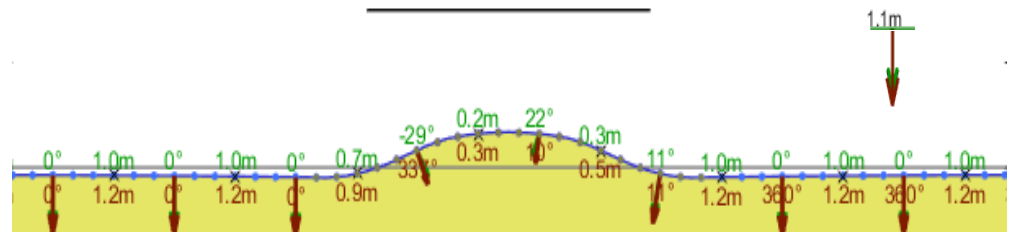


Figure 6-1 : Diffraction at an offshore breakwater

- **Groynes or port breakwaters**

These are cross-shore oriented structures which (partially) block or bypass alongshore sediment transport (Section 6.1.4) and affect the sheltered wave conditions in their shadow zone through diffraction and sheltering (Section 6.1.2). These structures have to cross the coast twice, and need to be defined in clockwise direction (Figure 6-2). So, it is sort of a 'staple'. It is possible to define curved, L-shaped or T-shaped groynes. The coastline will always stay connected to these groynes, as the coastline change routine always shifts the coastline along the perimeter of the structure. The coastline change along the groyne (Δs_{gr}) is approximately the same as the desired shore-normal coastline (Δn).

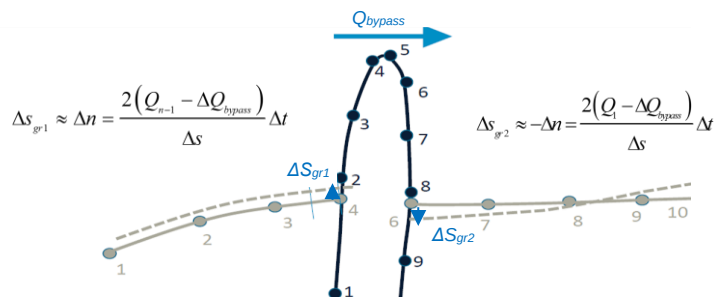


Figure 6-2 : Grid administration at groynes requiring a double crossing of the coastline

Tests with one (or more) groynes show that sediment accumulates at the updrift side of structures (Figure 6-3). The waves in the sheltered area of the groyne are computed using wave diffraction routines, resulting in a realistic erosion area on the leeward side of the groyne structure.

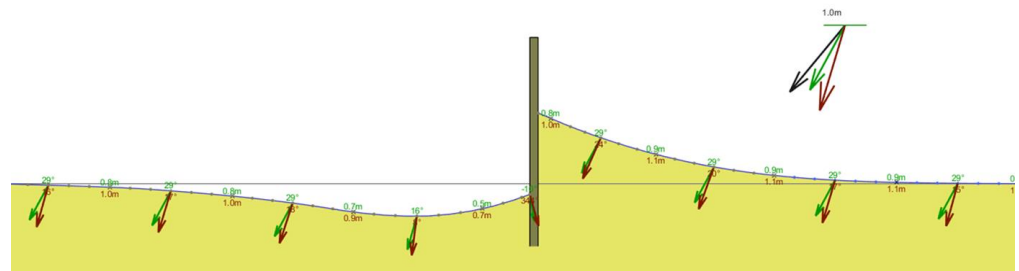


Figure 6-3 : Groyne scheme with diffraction and bypassing

A groyne may be placed at the start/end of a coastline section, where just one of the two legs of the groyne crosses the coastline. The coastline is virtually extended a few grid cells out to check when only one of the two legs of the groyne crosses the coastline. When the virtually extended coastline crosses the second leg of the structure, then it is recognized as a groyne at the boundary of the model. So, the beginning/end point of the coastline should be in-between both legs of the groyne. The model will then stick the coastline boundary to the groyne. It is recommended to combine this with a 'closed' boundary. The specified rate of transport at this boundary can be used as the bypassing of the groyne (see example in Figure 6-4).

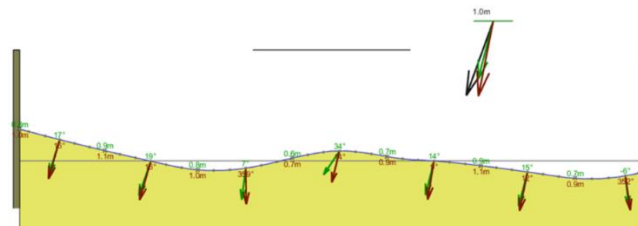


Figure 6-4 : Example model with groynes at the model boundaries.

- **Revetments / seawalls**

These are placed at the waterline and bypass sediment along their perimeter (Section 6.1.5). In addition, they can have an effect on transport rates when little sediment is present in front of these structures. Revetments can be either man-made coastal protections or natural rocky headlands.

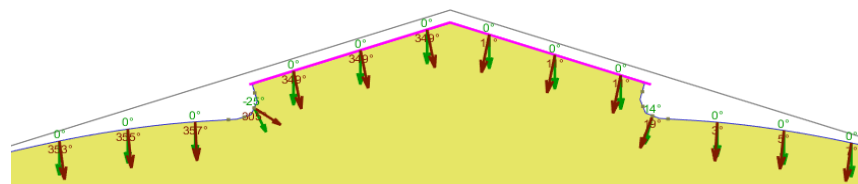


Figure 6-5 : Revetment with diffraction and bypassing

- **Hard erosion resistant layers**

Geological features may also limit erosion (Section 6.1.6). It can either be rock or coarse sediments with a negligibly low erosion rate. Alongshore transport rates depend on the width of the sediment layer in front of the hard layer.

Groynes and offshore breakwaters affect the wave conditions in their shadow zones as well as for the end points of revetments, which is discussed in Section 6.1.2. The bypassing of sediment at groynes is discussed in Section 6.1.4. Sea walls, rocky headlands and erosion resistant layers can create supply-limited situations where the sediment transport bypass is limited by the updrift sand supply, which are discussed in Sections 6.1.5 and 6.1.6.

6.1.1 Defining groynes, breakwaters and revetments

Groynes and offshore breakwaters can be switched on or off by setting the *struct* keyword to 0 or 1. For revetments the keyword *revet* should be set to 1, while the hard layers are switched on when the keyword *sedlim* is 1. If not set, then the model will try to detect the use of structures based on input of the coordinates. Structures are available throughout the whole model run. Re-starts of the model are needed to construct and/or decommission structures over time.

Structures can be defined by specifying an arbitrary number of polylines. This can be done either using a file reference or direct specification of x and y coordinates. For offshore breakwaters and groynes the keywords *xhard* and *yhard* can be used to provide vectors [Nx1] of the x and y coordinates, or alternatively a file can be referenced within the keyword *LDBstructures* which contains a [Nx2] matrix with the x and y coordinates. In both cases separate groynes/breakwaters should be separated by NaN's between the structures. The model will detect whether a structure is a groyne or an offshore breakwater depending on the number of crossings it has with the coastline. Offshore breakwaters should be located offshore at a somewhat similar distance from the coastline, and not be shading each other.

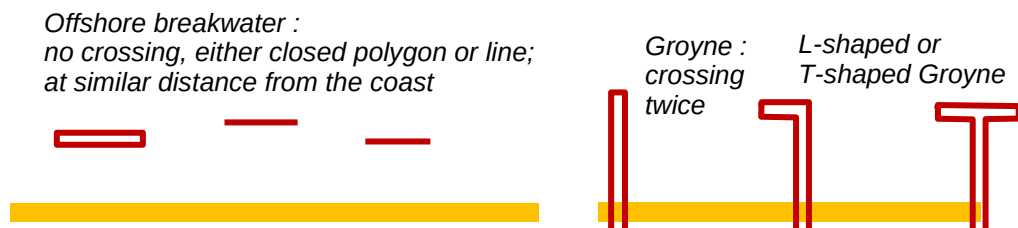


Figure 6-6 : Definition sketch of the offshore breakwater positioning

Revetment coordinates need to be specified either as [Nx1] vectors using the keywords *xrevet* and *yrevet*, or as a text-file reference in *LDBrevetments* (as a string of the filename). This textfile should then contain a [Nx2] table of the xy-points of the revetments. Multiple segments should also be separated by NaN's. The keyword *critwidth* sets the cross-shore width at which the transport starts to reduce.

The hard layers can be defined as input in a textfile, which is referenced in '*LDBsedlim*'. This file should contain a [Nx3] matrix with a column for the x-coordinates, y-coordinates and critical cross-shore width in front of the hard layer at which transports will start to linearly reduce to zero. Alternatively, the x and y coordinates can be specified as [Nx1] vectors with the keywords '*xsedlim*' and '*ysedlim*', and the critical cross-shore width with the keyword '*widthsedlim*'. Multiple elements should be separated with NaN's. The model will plot the 'virtual revetments' with sediment limitation as 'cyan' coloured lines.

6.1.2 Wave diffraction

Coastal structures create sheltered regions where diffracted waves determine the coastline change. Waves diffraction can therefore be accounted for in the model in the sheltered areas

of groynes, offshore breakwaters and at the ends of revetments. These diffracted waves are computed automatically for the affected (nearby) coastal areas when the diffraction is switched on by using *diffraction=1*. Note that the plots will show tiny grey dots at coastline points for grid cells where diffraction is evaluated.

The wave height reduction (k_d) at the sheltered coastline point depends on the re-orientation of the incoming wave angle (ω_r) as the diffracted wave components are now coming from the diffraction point. The directional spreading (*dirspr*) can be specified, which adjust the amount of diffraction.

$$H_d = k_d \cdot H_i \quad \text{wave height impact} \quad (48)$$

$$\varphi_d = \varphi_i + \omega_r \quad \text{diffracted wave angle} \quad (49)$$

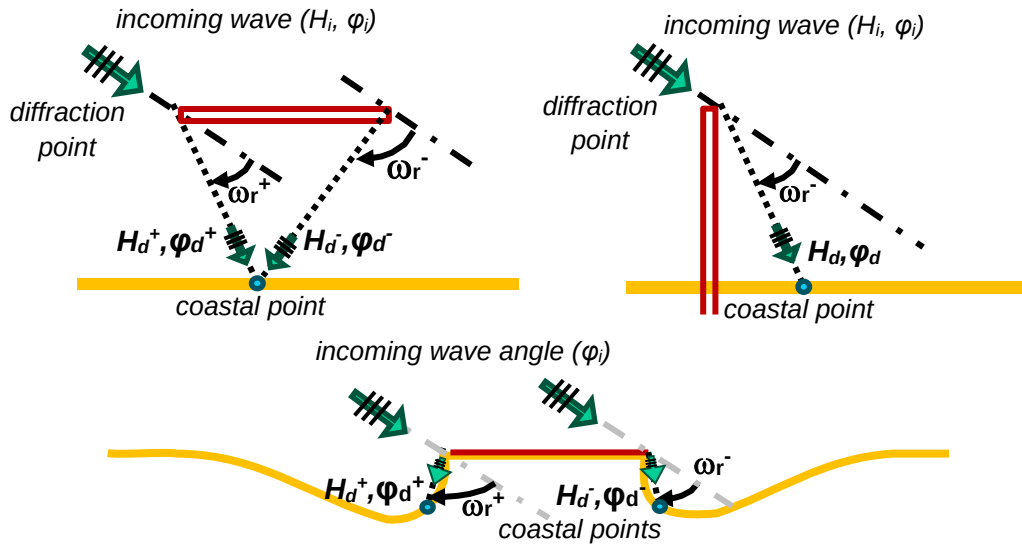


Figure 6-7 : Principle of wave diffraction at coastal structures

With a diffraction coefficient k_d , diffracted wave height H_d and incident wave height H_i . The diffracted wave angle is φ_d is the incoming wave angle φ_i plus a rotation component ω_r (see Figure 6-7). Note that some structures can have diffracted wave components from two sides (H_{d+} , ω_{r+} and H_{d-} , ω_{r-}). Keywords can be set to select the wave height impact of diffraction (*kdform*) and for the impact on wave angles (*wdform*). Two options are available :

- Kamphuis (2000) and Dabees & Kamphuis (1998) parameterization
This formulation is based on laboratory experiments with long-crested wave conditions. The wave height reduction is discretized for different ranges for the rotation of the wave angle at the diffraction point (ω). Note that the rotation of the wave angle at the diffraction point (ω) is used directly in the shaded zone for long-crested waves. For waves with directional spreading (*dirspr*) a probabilistic approach is used which weighs the contribution for evaluated directional sectors. The Kamphuis approach provides a relatively smooth spatial gradient in diffracted wave energy.

$$\begin{aligned} k_d &= 0.69 + 0.008 \omega_r & \text{for } 0^\circ > \omega_r > -90^\circ \\ k_d &= 0.71 + 0.37 \sin(\omega_r) & \text{for } 40^\circ > \omega_r > 0^\circ \\ k_d &= 0.83 + 0.17 \sin(\omega_r) & \text{for } 90^\circ > \omega_r > 40^\circ \\ \omega_r &= \omega \end{aligned} \quad \begin{aligned} kdform &= 'Kamphuis' \quad (50) \\ wdform &= 'Kamphuis' \quad (51) \end{aligned}$$

- Roelvink parameterization

This approach uses a parameterization of diffracted wave conditions behind an offshore breakwater from Xbeach non-hydrostatic simulations (see Figure 6-8). The wave angle change at the diffraction point (ω_r) determines the wave height reduction (k_d) of the diffracted waves. It is noted that the local wave angles only get modified when the rotation exceeds a critical wave angle rotation (ω_t) at the diffraction point. The corrected angle rotation ($\omega_{r,corr}$) is used to compute the wave angle at the diffraction point ($\phi_d = \phi_i + \omega_{r,corr}$). The degree of rotation also depends on a spreading factor of the waves ($rotfac$). The Roelvink approach creates more pronounced local spatial gradients in diffracted wave energy than the Kamphuis approach. The formulations of the Roelvink approach read as follows.

$$k_d = 1 - \exp\left(-\left(\frac{0.5}{\omega_*}\right)^4\right), \omega_* = \frac{\omega_r + 90}{180} \quad kdform = 'Roelvink' \quad (52)$$

$$\omega_{r,corr} = \begin{cases} rotfac * (\omega_r - \omega_t), & \omega_r \geq \omega_t \\ 0, & \omega_r < \omega_t \end{cases} \quad wdform = 'Roelvink' \quad (53)$$

The values for the parameters were derived from a fit on the simulation results. The default value for the rotation factor ($rotfac$) is 0.8 and the rotation correction (ω_t) has a default value of -20° , which is representative for long-crested waves with a spreading of 12° or less. For waves with a larger directional spreading (keyword *dirspr*) the ω_t is automatically adjusted to a larger value in the model (with up to -35° for a directional spreading of 32° or more).

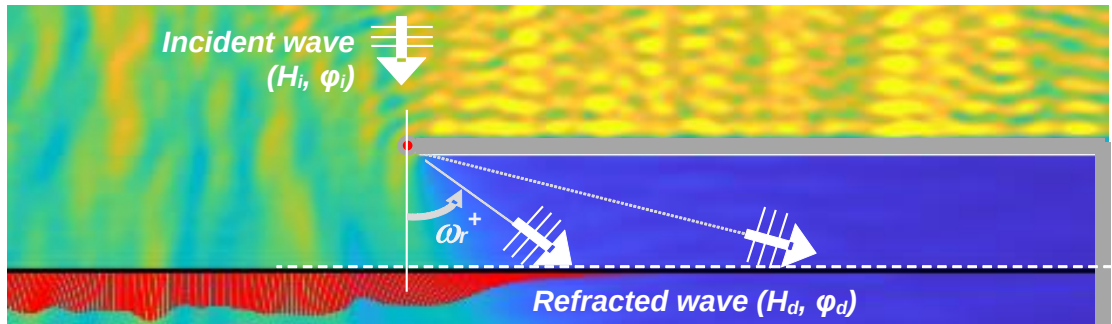


Figure 6-8 : Xbeach non-hydrostatic model ($H_s=1m$, $\phi_i = 360^\circ N$)

An offshore breakwater which is open at both sides requires the model to combine wave components of the diffracted waves from both sides of the structure using an addition of the wave energy vectors. The effect wave height reduction will be a combination of the energy of both vectors.

$$H_d = \sqrt{(k_d^- H_i^-)^2 + (k_d^+ H_i^+)^2} \quad (54)$$

The resultant direction from both components is obtained by finding the angle after the vectorial addition of the energy vectors of each of the components :

$$\omega_r = \tan^{-1} \left(\frac{(k_d^- H_i^-)^2 \sin \omega_r^- + (k_d^+ H_i^+)^2 \sin \omega_r^+}{(k_d^- H_i^-)^2 \cos \omega_r^- + (k_d^+ H_i^+)^2 \cos \omega_r^+} \right) \quad (55)$$

A point is in the influence area of a structure when $\omega_r^+ < \omega_{eps} - \omega_T$ and $\omega_r^- > \omega_T - \omega_{eps}$ for the given structure. The ω_{eps} is by default 90° . In between two structures, if a point is in the influence zone of both, the angle is computed in the same way, but the wave height is obtained by multiplying the k_d values from both adjacent structures. In an example for a single offshore breakwater, shown in Figure 6-9, we see that the wave height and direction are correctly approximated by combining the effects of the two diffraction points.

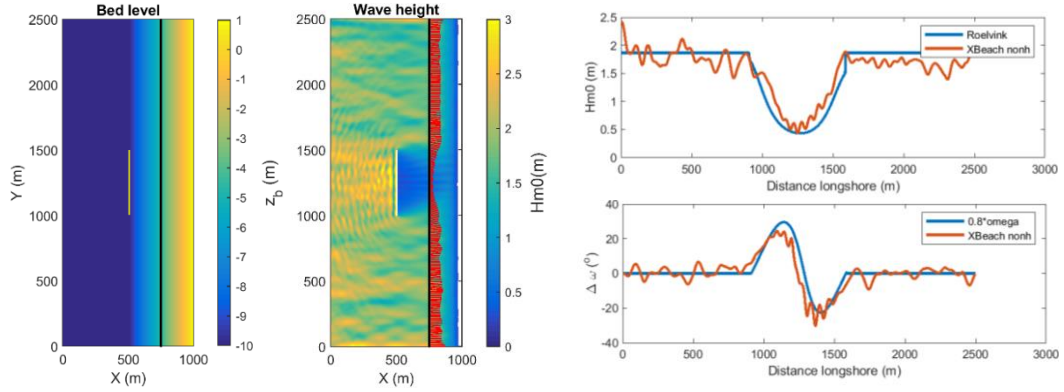


Figure 6-9 : A single offshore breakwater, at 500 m from shore, in uniform water depth of 10 m. Left panels: Model domain, bathymetry, reference line and computed wave height from Xbeach-nonh. Right panels: Variation in wave height (upper-right panel) and wave direction (lower-right panel) along the reference line, according to XBeach (red) and proposed formulation (blue).

An arbitrary number of structures can be combined, as can be seen for a case with 3 offshore breakwaters (Figure 6-10). For this purpose, it is very relevant that the ShorelineS model determines the influence area if each of the structures as described above. In addition, the model will prioritize the structure that enforces the largest rotation of the waves as the dominant structure when there is overlap, which can be the case for a very large groyne next to a small one, or when an offshore breakwater is located seaward of another offshore breakwater. In any case, it is not recommended to use complex configurations with overlapping structures. If waves are coming from a gap in-between two structures, then the components are combined first by multiplying their components (eq. 47 and 48).

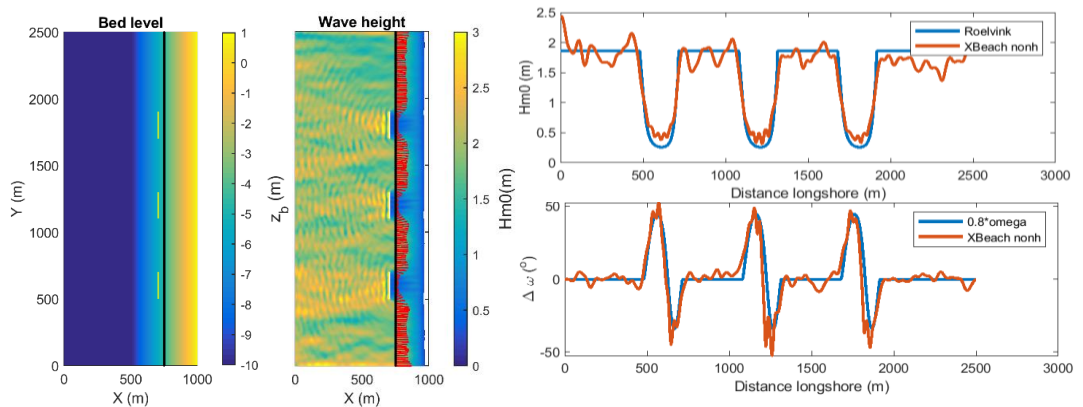


Figure 6-10 : Three small offshore breakwaters, at 500 m from shore, in uniform water depth of 10 m. Left panels: Model domain, bathymetry, reference line and computed wave height from Xbeach-nonh. Right panels: Variation in wave height (upper-right panel) and wave direction (lower-right panel) along the reference line, according to XBeach (red) and proposed formulation (blue).

The wave diffraction at groynes is very similar to that at offshore breakwaters, with the difference that one of the wave directions is blocked. Here an example is shown where the middle of the five structures is not an offshore breakwater, but a T-groyne which blocks the waves landward of it. The results show that the blockage is very effective (Figure 6-11).

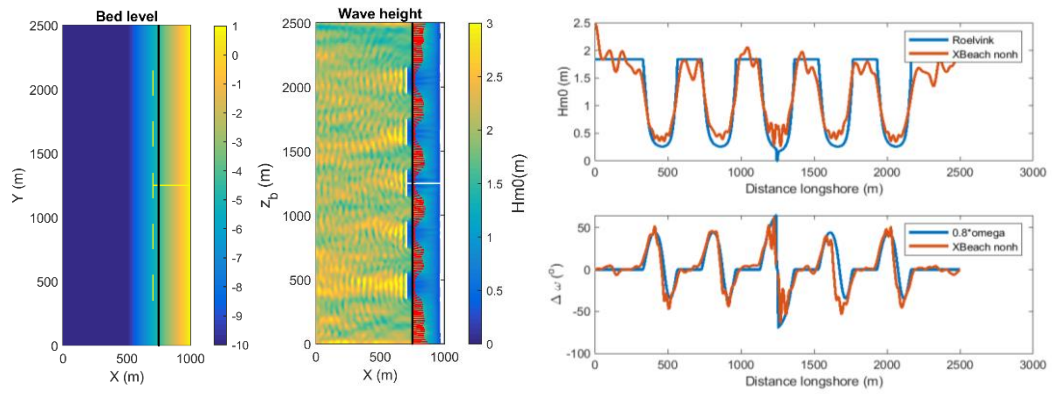


Figure 6-11 : Four offshore breakwaters with a T-groyne in the middle, at 500 m from shore, in uniform water depth of 10 m. Left panels: Model domain, bathymetry, reference line and computed wave height from XBeach-nonh. Right panels: Variation in wave height (upper-right panel) and wave direction (lower-right panel) along the reference line, according to XBeach (red) and proposed formulation (blue).

6.1.3 Wave transmission

Detached breakwaters, usually oriented parallel to the shoreline, are used around the world's coastlines to reduce wave attack and hence reduce erosion and/or flooding of the coast. Low-crested, detached breakwaters (LCB's) have their crest under water or are only slightly emergent for instance during low tide. These types of structures can be overpassed by incoming waves depending on the water level and design wave conditions, and only let part of the wave energy pass. The wave transmission calculation can be switched on with the keyword *transmission*. Wave transmission over groynes is not possible.

When switched on an additional transmitted wave component will be added for transmission at structures besides the components due to the wave diffraction at both lateral ends of the offshore breakwater. The effect of the diffracted waves is scaled back depending on the degree of wave transmission, scaling diffracted waves back to 0% when the transmission is 100%.

The module will compute the transmission coefficient (K_t) based on the user's inputs for the structure and climate conditions. This K_t represents the ratio between the wave height just shoreward of the structure transmitted wave height (H_t) and wave height just seaward of it incident wave height (H_i). The implemented routines in ShorelineS also re-orientate the waves at the coastal structure. The degree of rotation depends on the water depth at the structure, and can be switched on or off.

Characteristics of the structure

Transmissivity is a function of the structure's characteristics, wave parameters and the water level (see Figure 6-12). Relevant structure characteristics that are the crest height and width, front slope angle, breakwater roughness and permeability. For the wave transmission it is relevant to set the relevant parameters of the structure:

- The depth at which the offshore breakwater is specified is given in meters in *transmbwdepth*, with a positive value downward (e.g. 8 m).

- The crest height of the breakwater is specified in meters w.r.t. MSL in *transmcrestheight*, with a positive value upward, above MSL (so, -0.75 means 75 cm below MSL).
- The slope of the offshore breakwater is defined in *transmslope* as the ratio of the vertical change over the horizontal distance (e.g. a value of 1/3 is often used).
- The width of the breakwater crest is specified in meters in *transmcrestwidth*.
- The median stone size of the breakwater can be set with *transmd50* (default is 1 m).

If multiple structures are specified, then it is possible to use a different value for each of the structure by specifying a vector with the characteristics ([1xN] with length of the number of structures) for the above mentioned keywords.

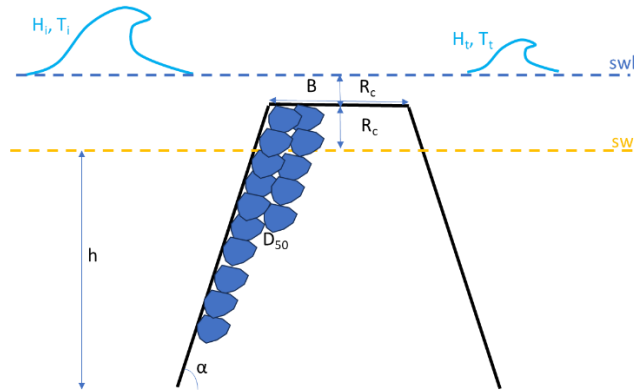


Figure 6-12: Definition sketch. Where H_i is incident significant wave height, R_c is the crest freeboard, B is the breakwater width, α is the seaward slope of the breakwater, h is the water depth of the breakwater base with respect to mean sea level, and D_{50} is the median size of the armour blocks.

Relevant hydrodynamic forcing conditions

Relevant wave characteristics are incident wave height, incident wave period and angle of the incoming waves. The tide plays an important role, as the freeboard over the submerged structures changes over the tidal cycle. Surge levels could potentially increase transmission during storms. Water-levels used for the computation of the wave transmission at the offshore breakwater water can be specified as a time-series in a text-file (*Watfile*). The water levels time series file has two columns [Nx2] with date/time in the first column (in 'yyyymmddHHMM') and waterlevel the second column (in meter relative to MSL). If water-levels are not specified, a fixed value of 0 meter is used for the water level.

Formulae for wave transmission

The newly implemented wave transmission over breakwaters can be performed using the approaches of 'd'Angremond et al. (1996), Van Gent et al. (2023) or Seabrook and Hall (1998).

The d'Angremond formula can be used with the keyword *transmform* = 'angr'. Based on laboratory tests, d'Angremond proposed expression for wave transmission (K_t) over rough, rubble mound breakwaters is:

$$K_t = -0.4 \frac{R_c}{H_i} + \left(\frac{B}{H_i} \right)^{-0.31} * (1 - e^{-0.5\xi}) * 0.64 \quad (56)$$

Where H_i is incident significant wave height, R_c is the crest freeboard, B is the breakwater width, λ is wave length, and $\xi = \tan(\alpha) / \sqrt{H_i/\lambda}$ is the Iribarren parameter. The transmission function is basically linear for situations with the non-dimensional freeboard (R_c/H_{m0}) between

-1 and +1.5. Please note that this formulation was derived for K_t values between 0.075 and 0.8, but is assumed the equation can work also in the range from 0 to 1. This equation is frequently cited in literature, robust, and is therefore selected as the default formulation in ShorelineS. However, it is recommended to also test with the van Gent et al. (2023) equation.

The most recent publication on transmission estimations over submerged breakwaters is the one of Van Gent et al. (2023), which is based on new laboratory measurements. It can be activated with the keyword *transmform* = 'gent'. For those measurements no upper limit of $K_t = 0.8$ is identified, but is extending up to $K_t = 1$, and seems for permeable, rubble mound structures overall quite similar to the one of d'Angremond. However, in the present tests for slightly submerged structures ($-1 < R_c/H_{m0} < 0$) the permeable structure showed less wave transmission than the impermeable structure, while the expressions by d'Angremond et al. (1996) predict the opposite for slightly submerged structures. For the more submerged structures ($-2.5 < R_c/H_{m0} < -1$) the expression of d'Angremond does not provide accurate estimates due to the upper limit (either $K_t = 0.8$ or $K_t = 1.0$).

The newly proposed expression by van Gent et al. (2023) for wave transmission over rubble mound breakwaters (also called “permeable breakwaters”) is:

$$K_t = c_1 \tanh \left(- \left(\frac{R_c}{H_i} + c_2 \left(\frac{B}{\lambda} \right)^{c_3} + c_4 \right) \right) + c_5 \quad (57)$$

Where H_i is incident significant wave height, R_c is the crest freeboard, B is the breakwater width, λ is wave length. Where the crest height is made non-dimensional using the wave height and the crest width is made non-dimensional using the wavelength based on the spectral wave period: $\lambda = 9.81/2\pi \cdot T_{m-1,0}^2$. The coefficients for rubble mound structures (permeable), are $c_1 = 0.43$, $c_2 = 3.1$, $c_3 = 0.75$, $c_4 = -0.25$, $c_5 = 0.5$. In Figure 6-13 their transmission function is shown as a function of the non-dimensional freeboard.

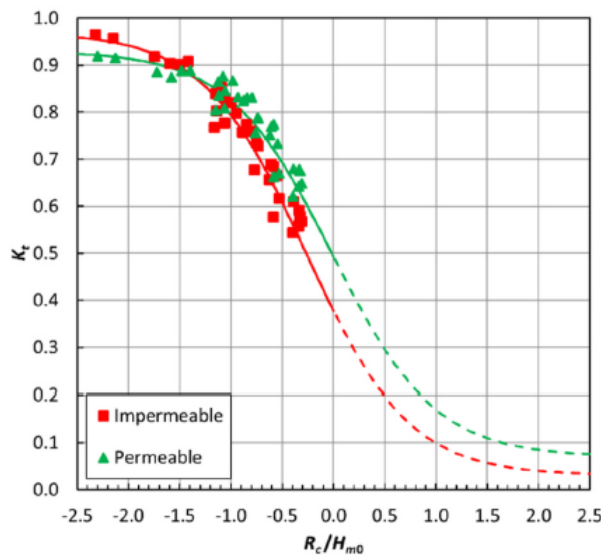


Figure 6-13: The included expression of rubble mound structures (permeable) in green. Source: van Gent et al. (2023).

The Seabrook & Hall formula can be activated with the keyword *transmform* = 'seabrhall'. Their proposed expression for wave transmission over rubble mound breakwaters (or as they call it “permeable breakwaters”) is:

$$K_t = 1 - \left(e^{-0.65 \frac{R_c}{H_i} - 1.09 \frac{H_i}{B}} + 0.047 \left(\frac{B R_c}{L D_{50}} \right) - 0.067 \left(\frac{R_c H_i}{B D_{50}} \right) \right) \quad (58)$$

Compared to the d'Angremond et al. and van Gent et al. equations, here the median size of the armouring layer (D_{50}) is incorporated as parameter. The equation is well bounded over the range of test data which is considered to be representative of typical design conditions. The equation does however become unbounded when B becomes very large or very small due to the size of the 3rd and 4th terms. Therefore, be cautious when applying the application outside of the following variable ranges:

$$0 \leq \frac{B R_c}{L D_{50}} \leq 7.08 \quad (59)$$

$$0 \leq \frac{R_c H_i}{B D_{50}} \leq 2.14 \quad (60)$$

Impact on wave angle

It is noted that by default the model will modify also the incoming wave angle of the transmitted wave such that it is more aligned (shore-normal to) the orientation of the submerged breakwater (default : *transmdir*=1). This 'influence of the structure on the wave-angle' scales linearly with the rate of transmission. For a transmission of 0% it uses a wave-angle that is perpendicular to the orientation of the offshore breakwater, while it uses fully the undisturbed wave-angle when the transmission is 100%. As an alternative, the undisturbed wave direction can be used always for the transmitted waves over the breakwater by setting *transmdir* to 0.

Implementation

The wave transmission routine in ShorelineS was tested for a theoretical case with an emerged and submerged breakwater. The breakwater has an alongshore length of 300 meter, a crest height of 0.75m below MSL, a crest width of 6 meter, a seaward slope of 1/3 and is located at a depth of 8 meter. The case shows that the emerged structure had a much stronger impact on the coast than the submerged structure (Figure 6-14).

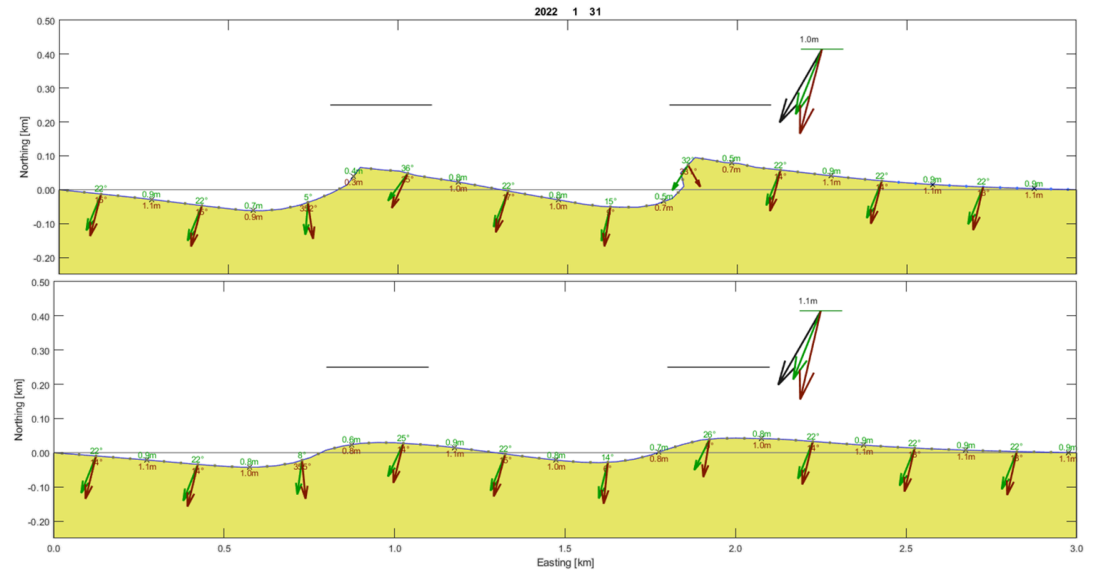


Figure 6-14: Coastline evolution behind an emerged (top panel) and submerged breakwater with partial wave transmission (bottom panel). The initial coastline is shown as the straight solid grey line.

In general, it is still recommended for users to calibrate the transmission function for an existing coastal structure with the different transmission equations before applying it to new structures.

Note that horizontal circulations due to submerged breakwaters as discussed in Ranasinghe and Turner (2006), which may have erosive or accretive effects, are not included in these formulations. This may be particularly important with short submerged breakwaters placed close to shore and under mostly perpendicular wave attack.

6.1.4 Bypassing at groynes

Alongshore bypassing of sediment can be of importance for ‘relatively short’ groynes that do not extend beyond the depth-of-closure, as well as for revetments. Attention is given to:

- 1) the bypass of sediment to the other side of the groyne and
- 2) the distribution of the bypassed sediment over the downdrift side of the structure.

Bypass of sediment to the other side of the groyne

The rate of bypass depends on the properties of the structure and the actual physical conditions (i.e. waves, tide and sediment). For the groynes, an approach is used to determine the degree of bypassing, which scales the transport based on a ratio of the depth at the tip of the structure (D_s) and the active depth of the longshore transport (D_{LT}). So, sand bypassing occurs at the seaward end of the groyne as long as D_s is less than D_{LT} .

$$\begin{aligned} QS_{bypass} &= \left(Fc - \frac{D_s}{D_{LT}} \right) QS_{updrift} & \text{if } D_s \leq D_{LT} \\ QS_{bypass} &= 0 & \text{if } D_s > D_{LT} \end{aligned} \quad (61)$$

With Fc typically being 1 (see below). Based on the assumption of equilibrium profile shape (Dean, 1977), the water depth at the structure's head D_s can be determined as:

$$D_s = A_p y_{str}^{2/3} \quad (62)$$

where y_{str} is the distance from the structure's head to the nearest point of the coastline [m] and A_p the sediment scale parameter [$m^{1/3}$], which depends on the median grain size (D_{50}), with $A_p = (1.04 + 0.086 \log(D_{50}))^2$. The depth of the active longshore transport (D_{LT}) is similar to the depth of the highest 1/10 waves at the updrift side of the structure (Hanson, 1989), and represents the time-dependent depth for longshore sediment transport, which is often less than closure depth D_c , and can be estimated as:

$$D_{LT} = \frac{A_w}{\gamma} (H_{1/3})_b \quad (63)$$

where $A_w = 1.27$, a factor that converts the 1/10 highest wave height to significant wave height [-]; γ is the breaker index, the ratio between wave height to wave depth at breaking line [-] and $(H_{1/3})_b$ is the significant wave height at the line of breaking [m]. In simulations with year-average wave conditions it may be better to scale up the active depth with the ratio of the yearly storm condition to the year-average wave condition. This parameter (A_w), which is by default 1.27 for time-series with varying conditions, should then be set to a higher value (e.g. $A_w=5$).

In the field the actual transport at groynes is often enhanced due to contraction of the flow (and tide) in front of the groyne. Consequently, the coastline seldom grows to the tip of the groyne, which can be the case in coastline models for cases with a dominant net transport direction.

An option is therefore built in to enhance the rate of sediment bypass at the groyne, which should be seen as a calibration factor that needs to be adjusted based on evidence at existing structures. The bypassfactor (F_c) can be specified by the user and is only applied for the transport from the updrift side of the groyne and when the coastline at the downdrift coastline is more landward than the coast at the updrift side of the structure. The F_c is typically between 1 and 1.4.

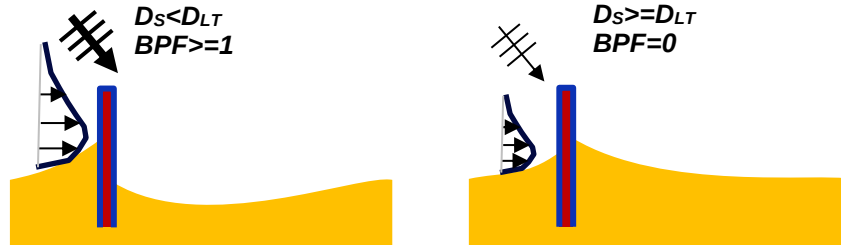


Figure 6-15 : Bypassing at a groyne for different wave conditions

Distribution of bypassed sediment at the downdrift side

Bypassed sediment is by default distributed evenly over the downdrift part of the groyne up to the point where the 'bypassed transport' is less than the 'unshadowed transport' would have been. It is possible to distribute the bypass closer to the groyne using the *bypassdistrpwr* with a value higher than 1. Two means a quadratic decay away from the groyne instead of linear.

$$Q_{s,i} = Q_{s0,i} + Q_{s,bypass} \cdot \left((1-i) / n_{bypass} \right)^{pwr} \quad (64)$$

With $Q_{s,i}$ as the transport in the shadowed zone with bypassed sediment, $Q_{s,i0}$ as the transport in the shadowed zone before bypass, $Q_{s,bypass}$ the bypass transport rate, n_{bypass} the number of grid cells in the shadow zone with a transport smaller than $Q_{s,bypass}$, i the relative grid index number with respect to the first point in the shadowzone downdrift of the groyne (always positive, starting at 0) and pwr the bypass distribution power.

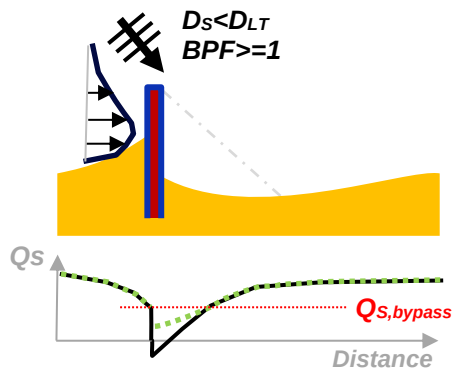


Figure 6-16 : Redistribution of bypassed sediment in the shadowing zone of a groyne

6.1.5 Revetment bypassing

Revetments preserve the local coastline and bypass sediment from one side to the other. This is in the model achieved by limiting sediment bypass along the revetment to the rate of the updrift supply. In case sediment is present in front of the revetment, the transport can increase only to the extent that the available sediment buffer allows. To avoid that the coastline retreats behind the revetment, while conserving mass, we limit the time-integrated flux at each transport point to the volume of the updrift supply and local sediment. The model will make a number of

sweeps along the revetment to ensure the bypass is correct, which can be set with the keyword *iterrev*.

$$Q_s \leq Q_{s,updrift} + \frac{\Delta s \cdot \Delta n_{rev} \cdot D_c}{2 \cdot \Delta t} \quad (65)$$

where Δs is the local grid cell size, D_c the active height of the profile height, Δt the time step and Δn_{rev} is the width of the beach in front of the revetment.

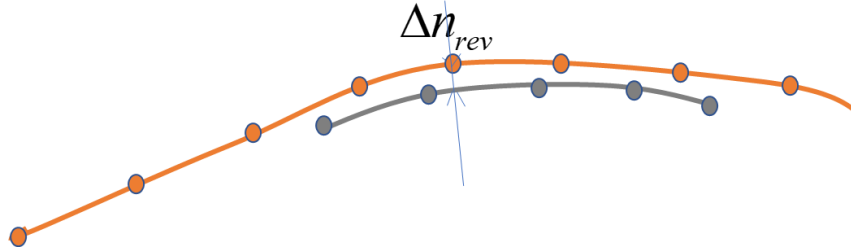


Figure 6-17 Definition of beach width (bottom panel) and example reduction of transport (top), where the dashed line is the original transport and the drawn line the modified transport rate.

Revetments also have a transition zone (or cross-shore width), wherein they linearly reduce transport rates when the width of the beach in front of the revetment (Δn_{rev}) is below a critical threshold (Δn_{crit}), which is illustrated in Figure 6-18. The keyword *critwidth* sets the cross-shore width at which the transport starts to reduce.

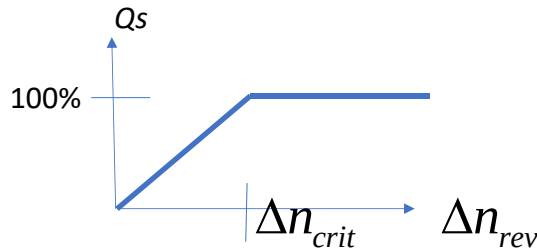


Figure 6-18 Decrease of longshore transport, Q_s , as function of the beach width.

Finally, to avoid gaps between the revetment and the coastline, the coastline is shifted to the revetment where necessary at the model start, i.e. when the revetment is initially seaward of the coastline.

6.1.6 Bypassing at hard layers

A hard layer can be seen as a virtual revetment, which blocks erosion and reduces sediment transport along a section of coastline. The already mentioned *xsedlim* and *ysedlim* or *LDBsedlim* keywords can be used for the input. The transport reduces linearly from 100%, when the active layer of sediment is wider than a critical/threshold width (*widthsedlim*). So, a transport of 0% is used when no sediment is present anymore in the region in front of the hard layer with limited sediment availability. This works in a similar way as for the revetments. It is also possible to specify a negative value for the critical width, which makes it possible to bypass sediment more easily along the hard area.

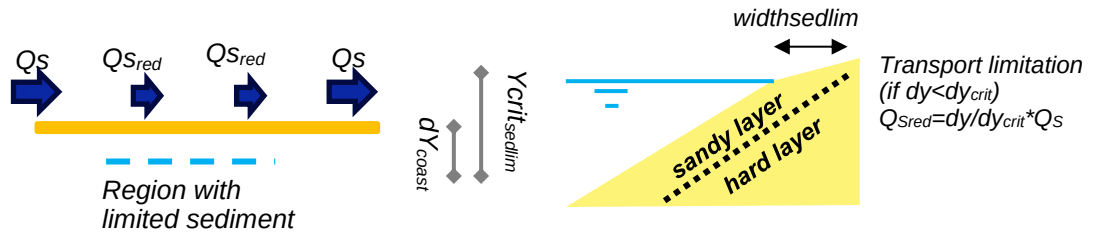


Figure 6-19: Reduced transport at a hard layers.

6.2 Nourishments

Two types of nourishments can be applied. The first is the regular beach nourishment, which is placed directly at the waterline. The second option is the shoreface nourishment, which is nourished on the middle shoreface, from where it gradually provides the beach/waterline with sediment.

6.2.1 Beach nourishments

Regular beach nourishments can be switched on or off by setting *nourish* (with 0 being off, and 1 being on). The nourishments then need to be prescribed in a file with '.nor' extension that is referenced in the keyword *LDBnourish*. This NOR-file contains a table with a line of information for each nourishment. Each nourishment should be defined as:

- x and y location of the begin of the nourishment (columns 1 and 2)
- x and y location of the end of the nourishment (columns 3 and 4)
- start and end time of nourishing in 'yyyymmdd' (columns 5 and 6)
- total volume of the nourishment in m^3 (column 7)

Alternatively, a polygon may be defined with a polygon file (*LDBnourish*='example.pol' with nourishment locations [Nx2] and multiple nourishments separated by NaN's). Furthermore, the rate of the nourishment (*nourratefile*='example_rate.txt' in m^3/yr), start and end date (*nourstartfile*='example_start.txt'; *nourendfile*='example_end.txt' in 'yyyy-mm-dd') need to be provided, with a line for each of the considered nourishments

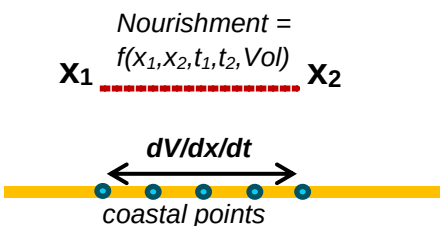


Figure 6-20: Distribution of sediment at nourishments.

This information is then internally converted into a shoreline accretion rate by dividing the total volume over the time period, the length of coastline and the profile height, D_c . By the same mechanism sediment discharged by a river can be distributed over a coastline section within a specified polygon.

It is noted that nourishment can also be used to prescribe sediment sinks (e.g. at beaches with steep tidal channel slopes) or to add sediment at places with bar-welding.

6.2.2 Shoreface nourishments

Shoreface nourishments are placed as relatively long (2-10 km) sand bodies in depths ranging from MSL -10 to -4 m. They simplify the nourishing process as the dredging vessels can navigate up to the shoreface nourishment location and quite rapidly unload the sediment. The volume is, however, not instantaneously available at the waterline, as it takes time for the sediment to be picked up by the waves and moved onshore. So, the sediment more slowly feeds the coast than for a beach nourishment. Consequently, a method is needed for the ShorelineS model which adequately describes the feeding of the coast by these shoreface nourishments. The shoreface nourishments can be switched on using $fnourish=1$. This involves two steps: 1) an assessment of the rate of feeding of shoreface nourishments in relation to their properties and environmental conditions for field data, for which a diffusion coefficient is estimated, and 2) a mechanism in the model that 'stores' the nourished sediment at the shoreface, and slowly feeds it to the beach using the diffusion coefficient.

Estimating a diffusion coefficient from shoreface nourishment data

A dataset of 19 shoreface nourishments along the Dutch coast was used to investigate the relation between a suitable diffusion K_{SFN} factor and wave and nourishment characteristics. For these 19 shoreface nourishments the volume evolution over the first few years is known (based on bed levels, one survey per year; see Huisman et al, 2019). An approximately linear trend was found in the volume decrease over the first few years. It should be noted that this linear trend does not necessarily also have to hold for the subsequent years.

The best-fit K_{SFN} -values for all nourishments, based on aggregated wave characteristics¹, were plotted against nourishment characteristics (length, depth and initial volume) and aggregated wave parameters (see Figure 6-21). The strongest correlation can be found with the initial nourishment volume. A larger shoreface nourishment has a lower K_{SFN} -value and a smaller decrease in nourishment volume over time.

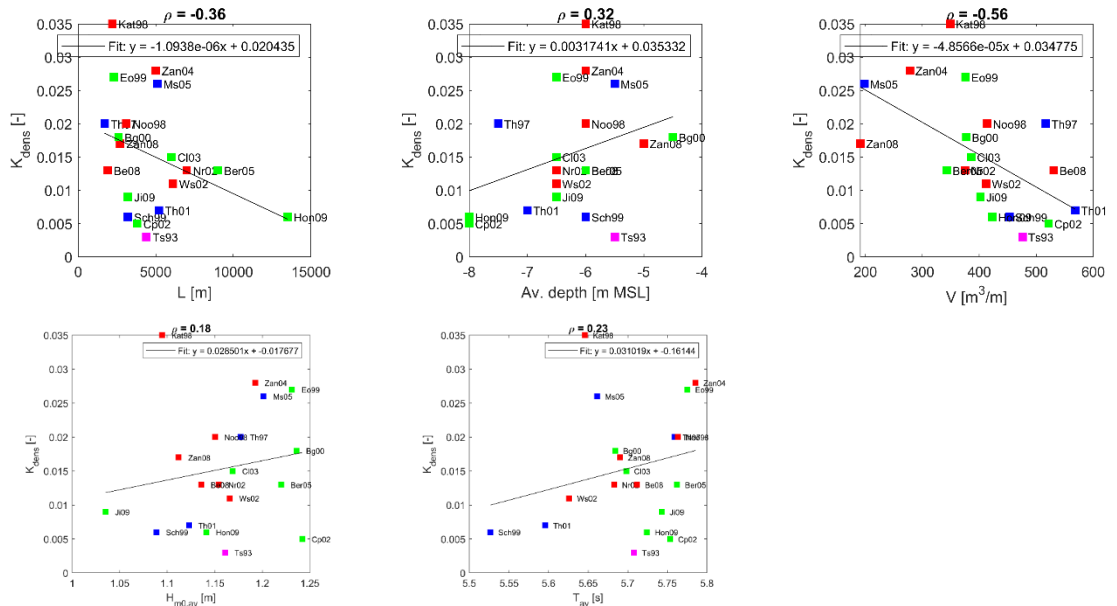


Figure 6-21: Relation of best-fit K_{SFN} -values for the 19 shoreface nourishments against nourishment characteristics (top panels : Length of the nourishment, Average Depth and Volume density) and wave characteristics (bottom panels : average wave height and average wave period).

¹ Using wave timeseries instead of aggregated wave parameters does not significantly alter the best-fit K-value.

Based on the fit between initial nourishment volume and the best-fit K_{SFN} -values for the 19 nourishments, a relation for K_{SFN} was derived:

$$K_{SFN} = \max(-4.86 * 10^{-5} * V_0 + 0.035, 0.001) \quad (66)$$

With V_0 the initial volume in [m³/m]. It is noted that the initial volume automatically also accounts for the water depth at which the nourishment was placed, which was found to be correlated. The K_{SFN} is always at least 1e-3. It is noted that this routine is only used when the diffusion coefficient of the shoreface nourishment is not prescribed by the user (with keyword K_{SFN}). And for cases outside the Netherlands it is advised to use a user-defined diffusion coefficient.

Computing the supply rate of the shoreface nourishment to the coast

The rate of supply of the shoreface nourishment to the coastline (q_{sfn}) is computed by the model using a diffusion coefficient. A formulation by Hallin et al (2019) is used to keep track of the nourished sediment. This formulation assumes the volume decreases exponentially and shore-normal incident waves.

$$q_{sfn} = \frac{V_{sfn}(1 - \exp(-K_{sfn} * \lambda * dt))}{dt} \quad (67)$$

$$\lambda = \lambda_0 * \left(\frac{H_0}{wT}\right)^{mb} \quad (68)$$

Where q_{sfn} is the onshore transport rate [m³/m/s], V_{sfn} is the volume of the shoreface nourishment [m³/m], K_{sfn} is the diffusion coefficient (0-1) [1/s], λ is the scaling factor for the supply rate [-], λ_0 as the calibration coefficient (0.56e-6), mb as a scaling power (-0.5), H_0 as the offshore wave height [m], w is the settling velocity [m/s] and T as the wave period [s]. The calibration coefficient has been derived from measured volume changes at 19 shoreface nourishments at the Dutch coast.

The remaining volume on the shoreface is kept track of for each shoreface nourishment at each timestep and for each coastal segment. The distribution of the supply of a shoreface nourishment over one or two coastal segments (e.g. two islands) is computed based on the length of coastline that is influenced by the nourishment, for which a projection is made of the lateral extents of the shoreface nourishments at the coast. In case the nourishment has been placed before the simulation start time (based on average wave conditions) then the initial volume is computed.

The computed onshore transport rate (q_{sfn}) from the shoreface nourishment is added to the formulation for coastline change as a source (in m³/m/yr). In this way the coastline change is affected by the supply of the shoreface nourishments. The onshore transport rate and remaining shoreface volume is saved.

$$\frac{\partial n}{\partial t} = - \frac{1}{D_c} \frac{\partial Q_s}{\partial s} - \frac{RSLR}{\tan \beta} + \frac{1}{D_c} \sum q_i \quad (69)$$

Implementing shoreface nourishments

Multiple shoreface nourishments can be specified, which can overlap both in time and in space. It should be noted that a shoreface nourishment can only stretch over two coastal segments. The characteristics of the shoreface nourishments need to be specified in a separate file with file-extension .fnor, the filename needs to be specified with `fnorfile=filename.fnor`. This .fnor-file contains 1 row per shoreface nourishment, with at each column:

- Start x-location [m of coordinate system]
- End x-location [m of coordinate system]
- Start y-location [m of coordinate system]
- End y-location [m of coordinate system]
- Alongshore length of the shoreface nourishment [m]
- Placement depth [m]
- Median grain size (d_{50}) [m]
- Placement moment tstart in [yyyymmdd]
- The total volume of the shoreface nourishment [m³]

The coefficient K_{sfn} can either be chosen by the user or automatically computed by ShorelineS. To force a diffusion coefficient the user needs to specify a K_{sfn} per shoreface nourishment (specified as [Nx1]) using the keyword *K*. The alternative is that the model automatically estimates a suitable diffusion coefficient by leaving the diffusion coefficient empty (i.e. $K=[]$). Then the volume, length, grainsize and depth are used to compute the *K*. This estimation is, however, based on a dataset of 19 shoreface nourishments in the Netherlands (Huisman, et al., 2019) and therefore not suitable for other locations.

7 Cross-shore interactions

7.1 Barrier or spit overwash

Ashton & Murray (2006) introduced the physical process of overwash by assuming a minimum barrier width such that sediment eroded from the seaward side is deposited on the landward side. By simultaneously retreating the seaward and landward sides of a section narrower than the specified critical width, the retreating section creates a longshore transport gradient that tends to fill it up; thus, the retreating helps maintain the width.

A similar concept was implemented in ShorelineS in a simple approach for treating the barrier width, using the keyword '*spitwidth*' with a default value of 50 meter. At each time step, the model checks the local barrier width at each point/node, measured in the incident wave direction. If the barrier is narrower than the critical width, then overwash occurs. The overwash process moves the landward point a distance equal to the difference between the actual width and the critical width. Such a distance is not allowed to exceed a given percentage (e.g. 10%) of the local spatial discretization distance of the grid per time step to avoid discretization artefacts. Then the model looks for the closest node on the seaward side to erode it by the same amount. Note that during oblique incident waves it is less likely that overwash takes place as the distance over the spit is larger in the direction of the obliquely incident wave, for which the waves at the point of breaking are used.

Sediment from the seaward face is moved to the backside of the barrier, but the deposition can yield a larger cross-shore extension than the erosion from the highly energetic seaward side. Different active heights can therefore be set on the seaward and landward side of the barrier, which accounts for the smaller depths and milder conditions in the lagoon side (see Ashton and Murray, 2006). The active height at the landward side is the combination of the berm height (*Bheight*) and the closure-depth at the considered side (respectively *Dsf* at the seaward side and *Dbb* at the backside). So, the erosion width at the seaward face of the barrier may result in a somewhat larger accretion width at the backside of the barrier. The rate of overwash is determined by a timescale factor (*OWtimescale* in years). The barrier width will exponentially develop towards the equilibrium width over the provided period.

7.2 Dune interaction

The dune dynamics module is based on the approach of Larson et al. (2016). The main processes that are considered are dune erosion due to wave attack, and wind-blown sand transport leading to dune growth. The sand transport equations are formulated based on relevant physics in combination with empirical observations and then validated towards laboratory and field data. The features that are modeled include dune height (from dune foot to crest), the locations of the seaward dune foot, beach width and shoreline position (Figure 7-1). A summary of the included formulations are given below.

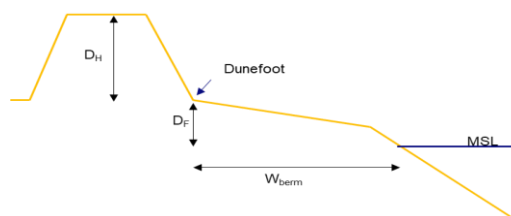


Figure 7-1: Specifications of the modeled features: dune height (*Dcelev*, dune crest elevation in ShorelineS code), dune foot height (*Dfelev* in ShorelineS code) and berm width (*Wberm* in ShorelineS code).

Wind-blown sand transport

Potential aeolian transport rate m_{WE} in $[kg\ m^{-1}\ s^{-1}]$ is defined following Lettau and Lettau (1978);

$$m_{WE} = K_w \sqrt{\frac{D_{50}}{D_{50r}}} \rho_a \frac{u_*^2}{g} (u_* - u_{*c}) \quad (70)$$

Where K_w is an empirical coefficient of 1.2 [-] based on Sherman et al. 2013. D_{50r} is the median reference particle size [m], ρ_a is the air density $[Kg\ m^{-3}]$ and u_* is the bed shear velocity $[ms^{-1}]$. The equilibrium transport rate per m width of the dune foot $[m^3\ s^{-1}\ m^{-1}]$ is,

$$q_{WE} = \frac{m_{WE}}{\rho_s} (1 - P) \quad (71)$$

with P as porosity, 0.4 [-]. The fetch length (F_L) is determined depending on the dry beach width and the wind direction (ϕ_{wloc} as wind angle to shore normal [°]). The combination of the fetch length and equilibrium transport rate can then be determined by,

$$q_{wind} = q_{WE} (1 - \exp(-\delta_{wind} F_L)) \quad (72)$$

where δ_{wind} is equal to 0.1, which is an empirical factor.

Wave-induced erosion

Dune erosion takes place only when the water-level (including run-up) reaches the dune foot. The total water level is the sum of tide level, surge level and wave runup height. Of these parameters the water level including the tide/surge should be provided as input (see Sections 4.2 to 4.4). The run-up is computed on the basis of the (offshore) wave height and beach slope (using berm width). You can select the run-up formulation of 'Stockdon', 'Larson' or 'Ghonim' with the keyword *runupform*.

- The Stockdon (2006) formulation includes the offshore wave height (H_0) and offshore wavelength (L_0) as well as the slope. It is the default formulation.

$$R = 1.1 \sqrt{H_0 L_0} * (0.35 * slope + \sqrt{0.563 * slope^2 + 0.004/2}) \quad (73)$$

- The Larson et al (2016) approach is independent of beach slope, which may lead to unrealistically high wave runup. It is useable for steep/narrow beaches, but not so much for mildly sloping beaches with considerable surge.

$$R = 0.158 \sqrt{H_0 L_0} \quad (74)$$

- The Ghonim (2019) formulation, is similar to the approach by Larson et al. It does, however, include also the beach slope, which makes it suitable also for wider beaches. It provides a run-up level that is similar to Stockdon.

$$R = slope \sqrt{H_0 L_0}, \quad (75)$$

The rate of erosion due to the wave attack can be calculated when the runup and storm surge levels are known.

$$q_{wave} = 4C_s \frac{(R' + SWL - D_F)^2}{T_H} \quad (76)$$

With C_s as impact coefficient of the foredunes against waves [-], D_F as the dune foot elevation which is defined with respect to MSL [m], SWL as still water level (tide + surge) [m] and T_H is the average wave period [s].

Coastline and dune foot change

After the wave attack and wind-driven transport have been determined, this is coupled back to the coastline position;

$$\frac{\partial n}{\partial t} = -\frac{1}{D_B + D_c} \left(\frac{\partial Q_s}{\partial s} - q_{wave} + q_{wind} \right) \quad (77)$$

where Q_s is longshore transport rate, D_B is the berm height, and D_c is the depth of closure. The coastline position change is denoted by $\partial n / \partial t$ [m/yr]. Subsequently the dune foot position changes as:

$$\frac{\partial n_d}{\partial t} = \frac{1}{D_h} (q_{wind} - q_{wave}) \quad (78)$$

In the schematization, dune foot elevation is assumed constant, while the beach slope can vary. However, when the slope is steeper than 1/15, the dune foot is lowered, so that the total water level is able to reach the dune more often, resulting in accelerated dune erosion for very narrow beaches.

Implementation in ShorelineS

To activate dune interaction, the keyword *dune* should be set to 1. The initial beach width, dune foot elevation and dune crest elevation need to be specified alongshore, in a file marked by *LDBdune=<filename>*. In this file, columns for x,y, beach width, dune foot elevation and dune crest elevation need to be specified. These properties are then interpolated alongshore in a way similar to nourishments or wave conditions. Output is generated for beach width W_{berm} , aeolian transport q_w and dune erosionflux q_s . The speed of dune erosion is regulated by a coefficient C_s with default value 5e-4.

The properties of the dunes then need to be specified. This can be done using a text-file with a number of typical dune properties along the coast, which is pointed to in *LDBdune*. This input file contains a [Nx5] matrix with at each line the local properties of the dunes:

- column 1 : x position of the specified dune properties
- column 2 : y position of the specified dune properties
- column 3 : berm width (or beach width) from waterline to dune foot (in meter)
- column 4 : height of the dune foot (w.r.t. MSL)
- column 5 : dune crest elevation (w.r.t. MSL)

It is noted that the difference between the 'dune crest height' and 'dune foot height' defines the active height of the dunes. So, both parameters should be defined positive upward. Note that in many cases the 'active height' of the coastline should be reduced when dunes are used in the model, as the dunes have their own active height.

Instead of the *LDBdune* keyword it is possible to specify above 5 parameters separately (if *LDBdune=""*). Then the model will check for *xdune*, *ydune*, *Wberm*, *Dfelev*, *Dcelev*. The following parameters (with default values) are relevant for dune erosion and dune growth:

$k_f=0.02$	Friction coefficient
$C_s=5e-4$	Dune erosion impact coefficient of the waves
$d_{50r}=2.5e-4$	Median reference grain size
$\rho_{\text{air}}=1.225$	Air density [kg/m ³]
$d_{\text{uneAw}}=0.1$	Coefficient (Bagnold, 1937)
$K_w=4.2$	Empirical coefficient (Sherman et al. 2013)
$k=0.41$	Von Karman's coefficient
$\text{segmaw}=0.1$	fraction of the fetch used for the aeolian transport computation

It is noted that dune growth is driven by wind forcing, which requires wind timeseries. Furthermore, dune erosion takes place only when the water level exceeds the dune foot level. This means that both the tide, the run-up, surge water levels should be defined.

As stated above, the run-up in the model is computed on the basis of the (offshore) wave height and beach slope (using berm width). You can select the run-up formulation with *runupform*. This can be:

- 'Stockdon'; (default preferred choice)
- 'Larson'; (for steep/narrow beaches and if no surge info available; gives higher runup)
- 'Ghonim'; (similar in formulation to Larson, but with beach width included, and therefore for wider beaches, run-up levels are more like Stockdon)

Test cases

A first test case evaluates an idealized north-south oriented coastline that is exposed to variable timeseries of wind, waves, tide and surge over a period of 22 years. This case demonstrates the variability of the coastline and dunes over these years. The initial active height of the underwater profile was set at 8 meter, the initial berm width at 50 m, the dunefoot at 3 meter above MSL and the dune top was set at 10 meter above MSL. An erosion coefficient (C_s) of $3e-3$ was used. Figure 7-2 shows the initial situation and forcing timeseries.

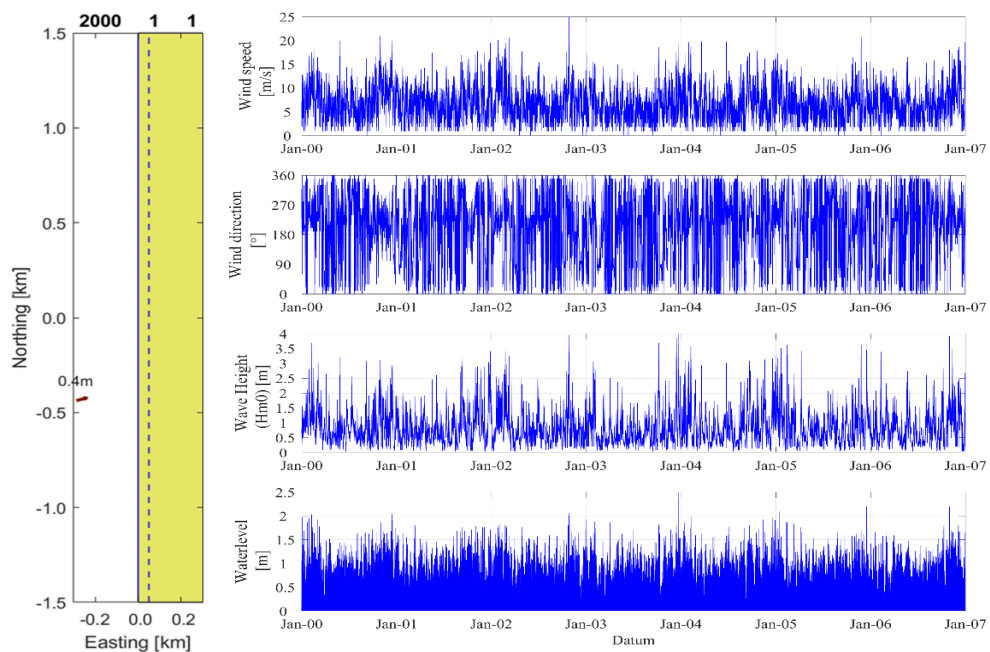


Figure 7-2: Initial coastline (solid line) and dunefoot position (dashed line) for a test case of the dune module with a realistic time-series of forcing conditions at Katwijk (right panels).

The test shows that the coastline and dunes are continuously exchanging sediment, with dune growth in milder periods and dune erosion in during sudden more extreme events.

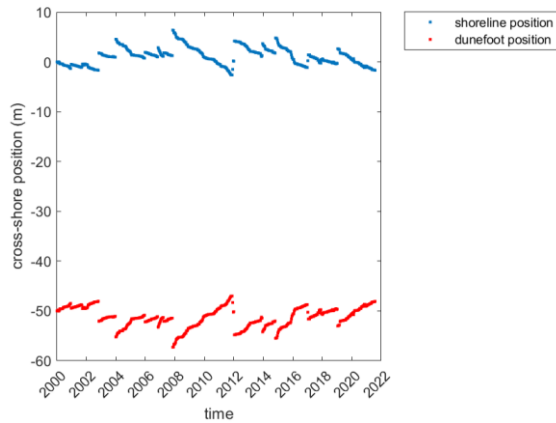


Figure 7-3: Resulting coastline and dunefoot positions over time in response to the variable forcing with surge.

A second theoretical example shows a curved coastline with and without a dune (see Figure 7-4). Over time the coastline is retreating on the north and south side due to longshore sediment transport. The eroded sand from the outer edges is transported to the central section of the coast. When switching on the dune module (right image in Figure 7-4) the dunes are reacting to the coastline change, on the north and south side the dunes are eroding, and consequently only a narrow beach remains. On the contrary, in the middle of the coast, where the coastline is accreting, this leads to a wider beach where windblown sand transport can occur, leading to accretion of the dunes.

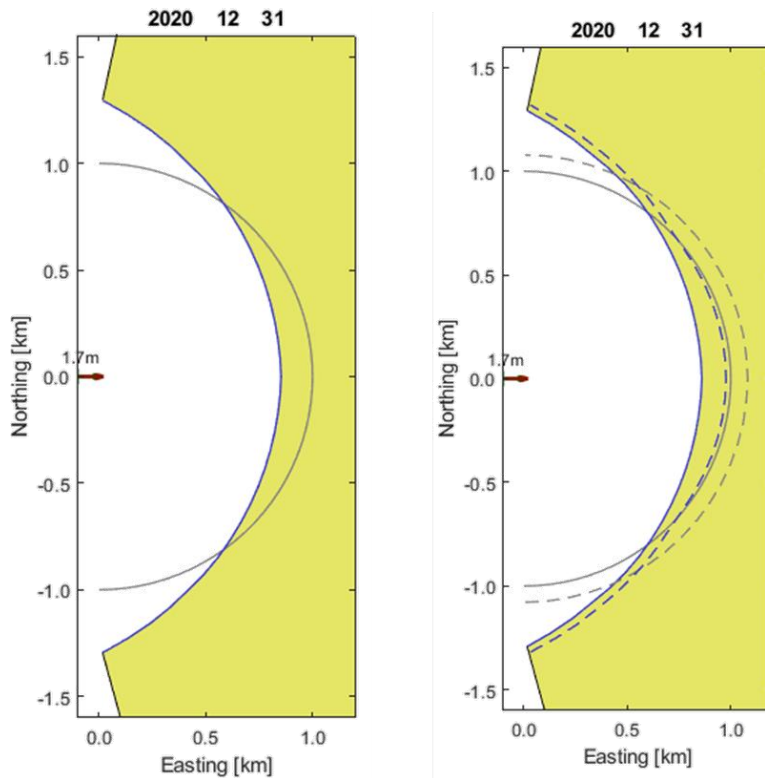


Figure 7-4: Test of a curved coastline, with (left) no dune taken into account, active height underwater profile of 12 m, and on the (right) with dunes taken into account, active height underwater profile is 8 m, dune height is 4 m. Initial coastline (solid line) and dunefoot (dashed line) position in grey, and end position in blue.

7.3 Cohesive / till dunes

Cohesive properties can be added to the dunes by specifying cohesive (till) properties of the dune. When used it means that 1) the erosion rate of the dunes is smaller (C_{still}) and 2) the eroded material is contributing only partially to the beach as a significant part of it is till/fines ($perctill$). The non-cohesive material of the dunes (i.e. non-cohesive fraction = $100\% - perctill$) will contribute to the growth of the coastal area when it is eroded.

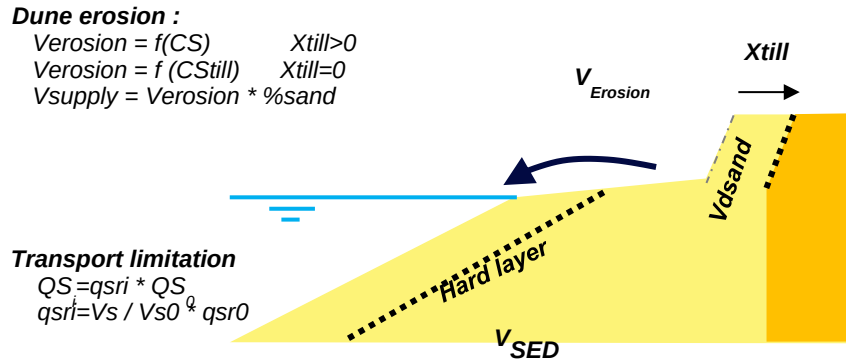


Figure 7-5 : Illustration of the sand layer (x_{till}) that can be present in front of a cohesive till dune.

A layer of sandy dune may be present in front of the cohesive/till dune. The layer can be specified as a 'cross-shore width' with respect to the initial coastline with the keyword x_{till} . Throughout the simulation the width of the sand-layer in front of the cohesive dune is tracked and stored in x_{till} . The model will first erode this non-cohesive sand layer of the dunes at the regular erosion rate (C_s) before erosion of the cohesive till takes place at a slower rate (C_{still}). Eolian transport is added to the sand layer (with limited width) that it is located in front of the cohesive dune layers, which means that the sand layer may also grow over time.

- The parameters of the till in the dunes can be specified either as fixed values, that are valid for the whole model domain, or as spatially varying parameters in the *LDBdune* file.
- The following parameters (with default value) are relevant for dune erosion/growth.
 - $C_{still}=5e-6$: Erosion coefficient of cohesive/till dunes
 - $x_{till}=[]$: Thickness of layer of sand in front of the till dunes. By default this is empty which means that an infinitely thick layer of sand is present. Setting 'xtill' to a value is required to activate the cohesive dunes in the model.
 - $perctill=80$: fraction of the dune sediment that is fines/till, which implicitly defines the remaining sediment as sand.
- Alternatively, spatially varying properties of the cohesive/till dunes can be specified instead, which overrules the standard settings. This can be done by adding extra columns [Nx1] to the *LDBdune* file which account for the cohesive properties of the dunes.
 - column 7 : erosion coefficient of the till (' C_{still} ')
 - column 8 : cross-shore width of the till (w.r.t. MSL)
 - column 9 : percentage of till in the cohesive dune sections (value between 0 and 100)

8 Interaction between coast sections

8.1 Merging and splitting

One of the advantages of the ShorelineS model is that it can simulate multiple coastal sections at the same time, and these sections can affect each other by shielding the waves. Small parts of the coast are allowed to split and migrate as the spits are growing and in some cases break up and migrate as a small island. Examples of the splitting procedure are shown in Figure 8-1 and Figure 8-2. Such splitting typically happens when the seaward side of a section erodes by more than the overwashing process allows for or when the latter is not activated. The numbering is indicated to show how the grid cell connections change after the splitting procedure: from one continuous coastline section to two separately numbered sections.

If two sections intersect, they may merge into one section as the simulation continues. Such merging typically happens due to shoreward migration or extension of a spit towards the mainland coast. Again, the numbering is included to indicate how the separate spit and mainland coast sections are now joined at the seaward side as a continuous coastline.

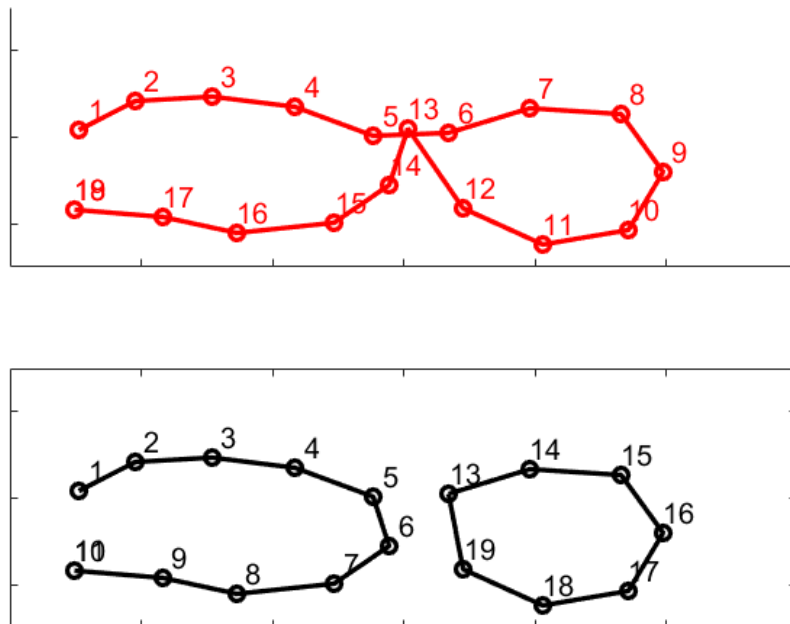


Figure 8-1 : Illustration of the splitting procedure; (A), before splitting and (B), after splitting.

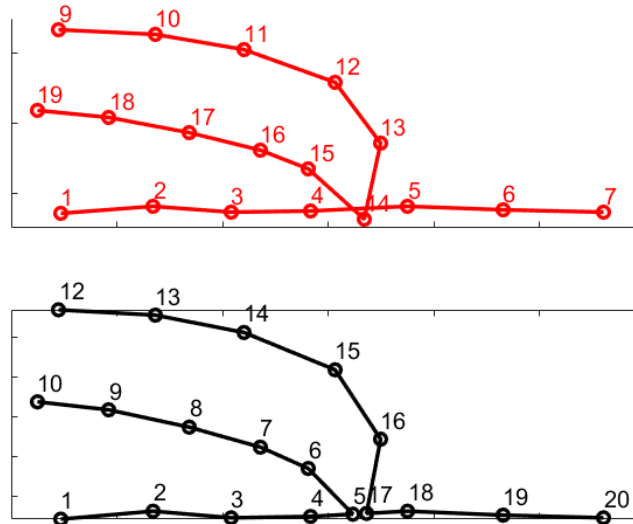


Figure 8-2 : Illustration of merging process; (A): two numbered sections before merging; (B): resulting sections and their numbering.

8.2 Channel migration

A function is available in the model which keeps a channel open. The model will force any sediment to the seaward side when the channel gets narrower than a critical width. An example is shown in Figure 8-3.

A channel path should be defined starting from a point inside the lagoon/river/channel and then going to a point outside the entrance. The x and y coordinates of this path can be specified either directly with [Nx1] vectors in the parameters *xrmc* and *yrmc*. Or alternatively, with a file in *LDBchannel*, which contains two columns respectively for the x-coordinates and y-coordinates of the channel path. Multiple segments can be separated by specifying NaN's.

The preferred width of the channel can be defined in *channelwidth*, which is specified as a [1xN] vector with a value for each of the defined channel pathways. The rate of transport of the channel can be scaled with the *channelfac* parameter. The value can vary between 0 and 1, with a default of 0.08.

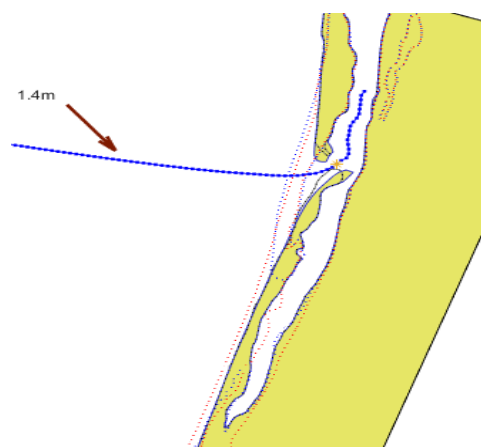


Figure 8-3 : Illustration of a channel that is defined for a coastal inlet in Senegal.

9 Post-processing

9.1 Output data

An output file ('output.mat') will be exported by the model to the specified output directory (*outputdir*). The time-interval for the output mat-file can be set in *storageinterval* which stores the interval in days. There are two methods of exporting data either using direct output (O-structure) or data that is re-interpolated to a grid (P-structure). Both are written to the 'Output.mat'.

The O-structure stores the raw information from the computation, which means that variables are defined at grids which may change in size over time. Also note that the O-structure stores the instantaneous data, and not the mean value since the previous timestep. This means the calculation of the cumulative transport can only be done when sufficient output timesteps are used.

The P-structure contains output data that is interpolated to a user specified (curvi)-linear grid. The grid is fixed in time and space. The input variable *xyout* can be defined in 2 ways:

- Specify two coordinates as [x1,y1,x2,y2]. A linear grid is made with *ds0* spacing. An additional line can be added to the matrix representing a new grid (e.g. [2x4]). So, multiple linear grids can be made.
- Specify a grid as a cell {} with [Nx2] grids of xy-coordinates. It is used 1 on 1 without interpolation of the gridsize. You can specify multiple grids by adding more cells. For example: {[5x2],[12x2],...}

The model will make a projection of the 'coastline position', 'wave conditions' and 'transports' on this grid. The exported coastline positions (*xc* and *yc*) provide the crossing points of the coastline in *ShorelineS*, at the output-moment, with the 'normal' of the gridline at each output grid cell. Also, the distance from the output grid to the crossing point of the 'normal' with the coastline will be stored (*zg*). When applying multiple grids, the P structure will have a dimension of P(gridnumber). The wave conditions (*Hs*, *Tp* and *PHI*) and sediment transport (*QS* and *QSmax*) rates in the P-structure are averaged over the period of the output timestep. So, it is not providing the instantaneous information like the O-structure.

It is noted that small output timesteps may result in a slowdown of the model during the run, which relates to writing the output data. In the future a file system will be introduced that can incrementally store the data throughout the simulation.

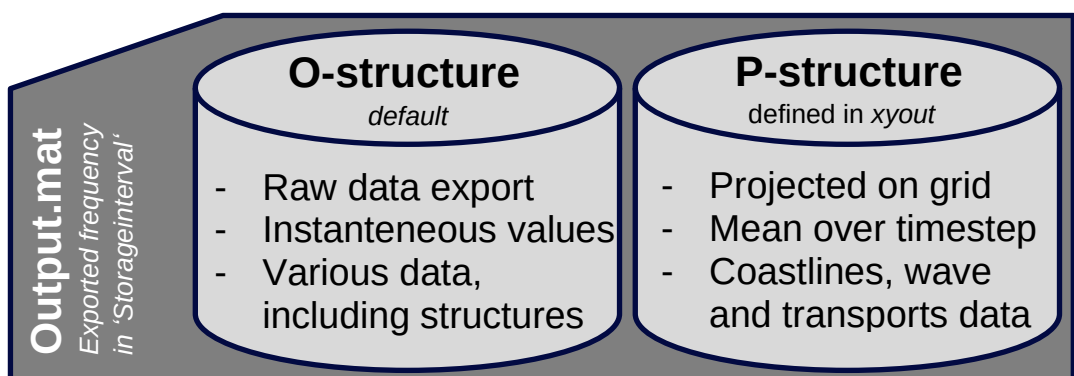


Figure 9-1 : Schematic of data in output.mat

9.2 Visualization of coastline and wave conditions

By default the average waves as prescribed on the model are plotted (at all defined locations) with the offshore wave (black arrow), nearshore wave (green arrow) and wave at point of breaking (brown arrow).

Showing local wave height and direction information

The *plotHS* or *plotDIR* keywords can be used to show more detail of the computed nearshore wave heights and/or wave directions along the coastline (see Figure 9-2). For example, *plotHS=6* will show you a value and arrow of the wave height at every 6th grid cell. While the *plotDIR=6* will show you an arrow of the waves at the nearshore depth (*tdp*) and point of breaking (*br*) respectively with a green and brown arrow. Similarly, the *plotQS* will show the transport rates along the coast at every n-th point, but with a slight offset.

With *plotUPW=1* it is possible to plot the locations where the high-angle correction is applied, which show up as green and red squares.

The background fill of the coast can be switched on or off using *usefill* (on=1, off=0). For complex coasts the *usefillpoints* can be set to a value larger than 0 to obtain a nice landfill behind the specified open coast, which creates the set number of points landward of the specified coastline. If *usefillpoints* with a value larger than 2 is used then a landpolygon will be created that curves with the coastline, which is useful for complex coastal shapes.

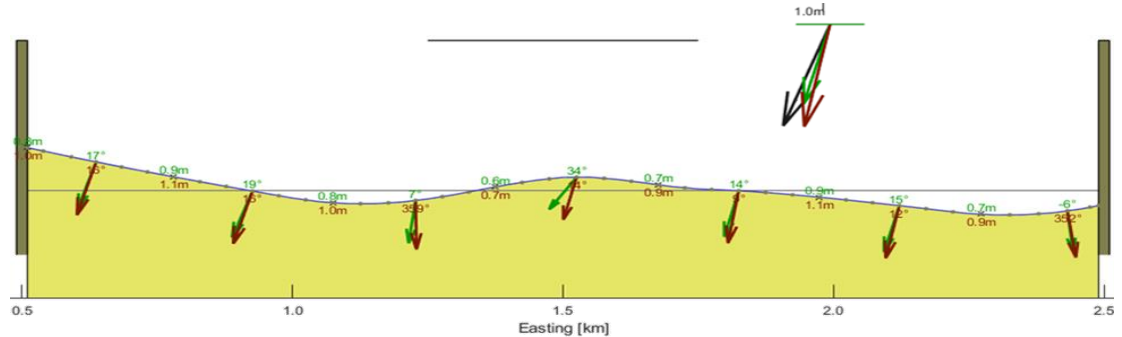


Figure 9-2 : Plot of a model with a beach scheme showing the applied offshore (wave) conditions and the local wave directions at the coast (for the nearshore in green and poin-of-breaking in red-brown).

Cross-shore plots with dune position

The model can make plots of the coastline and dune foot positions (i.e. cross-shore) at cross-sections that are specified in *xyprofiles*. The syntax is : *xyprofiles*=[x1,y1; x2,y2; xi,yi; ...]

Time interval for plotting and storing plots

The interval of plotting to screen can be set with *plotinterval* which contains the interval in number of timesteps (so, *plotinterval=2* means every two timesteps). A time-interval can be set for exporting the figures (*fignryear*) which has the number of occasions per year that a figure is written to file. Note that this cannot be more often than the *plotinterval*.

Spatial limits

XY-limits of the plot can be set with *xlimits* and *ylimits* ([1x2] of min/max coordinate in m). The position of the wave arrows and its scale can be set with *XYwave* ([1x2] coordinates). An offset can be extracted from the x and y-coordinates in the plot with *XYoffset* [1x2].

Output location

The output dir of the plots and the data is placed in *outputdir* (e.g. 'Output'). With *fastplot=1* keyword we can create debug plots, which are made very quickly. While more detailed/slower plots can be made using *fastplot=0*. In order to speed up the computation it may be decided to make the plot invisible by setting *plotvisible=0*. The plots will still be written to jpg-files in the output directory.

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A Code structure

The ShorelineS model is implemented in Matlab. The flow diagram of the model is depicted in Figure 2. In the following we will describe the procedure point by point.



Figure A-3. Flow diagram of the ShorelineS model.

B Overview of model input parameters

Table with an overview of the model input parameters and their default values.

Parameters	Default	Description
Wave conditions		
hso	1	wave height [m]
tper	6	peak wave period [s]
phiw0	330	deep water wave angle in degrees [°N]
spread	90	wave spreading of the wave climate [°] (i.e. variation over time)
dirspr	12	directional spreading of the wave [°] (i.e. 2D wave spectrum for diffraction at structures)
wvcfile	"	wave time-series <-leave empty to use wave parameters ('S.hso', 'S.phiw0' and 'S.spread')
ddeep	25	water depth the location of wave climate, corresponding with S.hso [m]
dnearshore	8	water depth at the 'dynamic boundary', corresponding with S.phif [m]
randomseed	-1	Seed to generate randomized series (any number larger than 0 gives a unique random seed)
interpolationmethod	'weighted_distance'	Method for interpolating stations along coast ('alongshore_mapping' / 'weighted_distance')
coastline definition		
ldbcoastline	"	file with initial coastline shape with [Nx2], or directly [Nx2] data
xmc	"	x-coordinates of coastline [Nx1], with NaNs separating the elements
ymc	"	y-coordinates of coastline [Nx1], with NaNs separating the elements
gisconvention	0	switch to swap coastline definition according to GIS convention (i.e. counter clockwise)
ds0	100	initial space step [m]
griddingmethod	2	method for regenerating the grid (1: only splitting and merging, 2: uniform grid regeneration)
d	10	active profile height [m]
phif	[]	Orientation of the foreshore [°N] (in degrees) <- only relevant for 'KAMP', 'MILH' or 'VR14'
maxangle	60	maximum coastline re-orientation between individual grid cells
preserveorientation	0	Switch forcing the model to preserve the original orientation of all coastal transects (0/1)
transport parameters		
trform	'CERC'	switch for transport formulation (e.g. S.trform='CERC', 'KAMP', 'MILH' or 'VR14')
b	1.00E+06	CERC : coeff in simple cerc formula
qscal	1	calibration factor of the transport (works for all transport formulas)
d50	2.00E-04	median grain diameter [m] (KAMP & MILH & VR14)
d90	3.00E-04	90th percentile grain diameter [m] (KAMP & MILH & VR14)
porosity	0.4	S.porosity (typically 0.4) [-] (KAMP & MILH & VR14)
tanbeta	0.03	mean bed slope [ratio 1/slope] (KAMP & MILH & VR14)
rhos	2650	density of sand [kg/m3] (KAMP & MILH & VR14)
rhow	1025	density of water [kg/m3] (KAMP & MILH & VR14)
g	9.81	gravitational acceleration [m2/s] (KAMP & MILH & VR14)
alpha	1.8	calibration factor point of breaking (S.alpha = 1.8 for Egmond data) (KAMP & MILH & VR14)
gamma	0.72	breaking coefficient (Hs/h) with 5% breaking waves (KAMP & MILH & VR14)
pswell	20	Percentage swell (between 0 - 100) [-] (VR14)
ks	0.05	roughness height [m]
hclosure	8	depth-of-closure [m]

cf	0.0023	roughness factor [-] used in transport formulation (TIDEPFROF)
n	0.02	manning roughness coefficient [$\text{s/m}^{1/3}$], if not empty is used instead of the 'cf' (TIDEPFROF)
acal	0.2	calibration coefficient for Soulsby Van Rijn [-]
hmin	0.1	minimum depth [m] used in transport formulation (TIDEPFROF)
sphimax	[]	switch for bypassing computation of critical coastline angle (e.g. at sphimax=42 or 'auto')
relaxationlength	[]	length over which transport decelerates in meters
suppresshighangle	0	switch 0/1 for truncing high-angle instabilities by limiting angles to the critical angle
time steps & numerical		
tc	1	switch for using adaptive time step (0=off), and fraction of automatic timestep (0 to 1)
dt	0	value of a fixed timestep for the coastline processes [fraction of a year] when S.tc=0
dtdune	[]	value of a fixed timestep for dune process [fraction of a year]
reftime	'2020-01-01'	reference/start time of the simulation ('yyyy-mm-dd')
endofsimulation	'2040-01-01'	the end time of the simulation ('yyyy-mm-dd')
twopoints	1	number of downdrift pointsover which high-angle transport correction distributes sediment
smoothfac	0	smoothing factor used to re-arrange grid every timestep (only griddingmethod=1) (0 to 0.1)
smoothrefrac	0	smoothing fraction of the coastline orientation for refraction (PHIcs) (0 to 1).
boundary condition		
boundaryconditionstart	'Fixed'	left boundary condition (either 'Fixed', 'Angleconstant', 'Periodic' or {'Closed',-6000} etc)
boundaryconditionend	'Fixed'	right boundary condition (either 'Fixed', 'Angleconstant', 'Periodic' or {'Closed',-6000} etc)
cyclic	0	switch for cyclic boundary conditions (e.g. for mud)
climate change impact		
ccslr	[]	rate of sea level (m/yr) rist or a table with [time1, sealevel1; ...]
cchs	[]	increase of the wave height (fraction/yr) or a table with [time1, HS increase factor; ...]
ccdir	[]	change in wave direction ($^{\circ}$ /yr) or a table with [time1, wave direction change; ...].
structures		
struct	1	switch for using hard structures (0/1)
ldbstructures	"	file with hard structures (with [Nx2]) / or [Nx2] matrix
xhard	[]	x-coordinates of hard structures ([1xN]) seperated by nans
yhard	[]	y-coordinates of hard structures ([1xN]) seperated by nans
structtype	{}	type of structures
aw	1.27	factor for determining depth of closure at bypassing groyne (i.e. when a time series is used)
awfixedhs	5	factor for determining depth of closure at bypassing groyne for a representative Hs.
bypassconfrac	1	scaling factor for the transport bypass at a groyne as a result of contraction of the flow (≥ 1)
bypassdistpwr	1	distribution of bypassed sediment of groynes over the shadow zone (1 to 2)
submerged	0	switch (0/1) to use submerged groynes (experimental feature)
groinelev	[]	elevation of submerged groynes (option to prescribe the vertical level of the groynes)
wave transmission over submerged breakwater		
transmission	0	switch to allow for wave transmission over breakwaters (0/1)
transmform	'angr'	transmission formula (d'Angremond (1996), Van Gent (2023) or Seabrook and Hall (1998))
transmdir	1	switch to adapt wave direction after transmission over breakwater (1=on, 0=off)
transmfile	"	file with characteristics of submergable breakwaters [Nx2]
transmbwdepth	[]	depth at which breakwater is positioned [m] Positive downward.
transmcrestheight	[]	breakwater height [m] w.r.t. MSL Positive upward, above MSL.
transmslope	[]	breakwater slope [-]
transmcrestwidth	[]	breakwater crest width [m]

transmd50	1	D50 of armour layer, needed for Seabrook and Hall.
permeable structures		
perm	0	switch for using permeable structures (0/1)
ldbpermeable	"	file with permeable structures ([Nx2])
xperm	[]	x-coordinates of permeable structures ([1xN]) seperated by nans
yperm	[]	y-coordinates of permeable structures ([1xN]) seperated by nans
wavetransm	[1]	wave transmission at permeable structures
revetments		
revet	1	switch for using revetments (0/1)
ldbrevetments	"	LDB with hard structures ([Nx2])
xrevet	[]	x-coordinates of revetments ([1xN]) seperated by nans
yrevet	[]	y-coordinates of revetments ([1xN]) seperated by nans
iterrev	5	iterations used to determine bypassing along a revetment
critwidth	5	critical width for determing coastal width in front of revetments.
wave diffraction		
diffraction	0	switch for using wave diffraction (0/1)
wdform	'Roelvink'	wave diffraction approach to treat angles (Roelvink(default) or Kamphuis/Hurst).
kdform	'Kamphuis'	computation of kd according to 'Kamphuis' or 'Roelvink' analytical approx.
rotfac	1.5	factor determining the rotation of waves due to diffraction
diffdist	[]	maximum distance of structures to coastline with diffraction (limits influence area)
nourishments		
nourish	0	switch for using nourishments (0, 1 or 2) -> 2 is common method used
growth	1	calibration of nourishment growth rate
norfile	"	file with nourishments (recommended method) with table of data [Mx7]
ldbnourish	"	LDB with polygons of nourishment locations ([Nx2]) (not used if .NOR file is available)
nourratefile	"	nourishment rates for each nourishment polygons (not used if .NOR file is available)
nourstartfile	"	nourishment start dates for each nourishment polygons (not used if .NOR file is available)
nourendfile	"	nourishment end dates for each nourishment polygons (not used if .NOR file is available)
nourrate	100	rate of nourishing (not used if .NOR file is available)
shoreface nourishments		
fnourish	0	switch for shoreface nourishments (0/1)
fnorfile	"	shoreface nourishment input file
sal	35	salinity of the water [ppm salt]
temp	10	temperature of the water [degrees celsius]
mb	-0.5	coefficient 1 for shoreface nourishments supply over time
labda0	5.60E-07	coefficient 2 for shoreface nourishments supply over time
k	[]	coefficient 3 for shoreface nourishments supply over time
sources and Sinks		
sourcessinks	"	sources and sinks rates
ssfile	"	sources and sinks file
aeolian transport to the dunes		
dune	0	switch for using estimate dune evolution (0/1)
ldbldune	"	filename with dune parameters (.dun file) [Nx5] with xdune,ydune,wberm,dfelev,dcelev
kf	0.02	friction coefficient
cs	5.00E-04	erosion coefficient of the dunes during storms, which scales the rate of erosion

cstill	5.00E-06	erosion coefficient of the dunes during storms for dunes with consolidated till layers
xtill	[]	thickness of layer with sand at the seaward side of the dunes in front of a till layer
percstill	80	percentage of till in cohesive dune layer. [Nx3] table with x,y,percstill per location
d50r	2.50E-04	median reference grain size
rhoa	1.225	air density []
duneaw	0.1	coefficient (Bagnold, 1937)
kw	4.2	empirical coefficient (Sherman et al. 2013)
k	0.41	von Karman's coefficient
segmaw	0.1	empirical factor used for scaling impact of the fetch length
maxslope	jan-15	the maximum slope angle (1:slope) with default of 1:20.
aoverwash	3	rate of overwash
xdune	0	x coords where wberm and dfelev are given (only used if ldbdune is empty)
ydune	0	y coords where wberm and dfelev are given (only used if ldbdune is empty)
wberm	50	berm width (m) for fetch of the eolian transport & the run-up (only used if ldbdune is empty)
dfelev	3	dune foot elevation w.r.t. MSL (m) (only used if ldbdune is empty)
dcelev	8	dune crest elevation w.r.t. MSL (m) (only used if ldbdune is empty)
wind conditions		
wndfile	"	wind time-series filename <-leave empty to use 'S.uz', 'S.phiwnd0' and 'S.spread'
Cd	0.002	wind drag coefficient (-)
uz	"	wind velocity at z (m)
z	10	elevation of measured wind data
phiwnd0	330	wind angle [degN]
still water levels and runup		
runupform	'Stockdon'	switch for run-up formulation (default 'Stockdon', or 'Ghonim', or 'Larson')
runupfactor	1	tuning factor for runup (linear)
watfile	"	Water level time series file [Nx2] date/time in 'yyyymmddHHMM' and waterlevel wrt MSL
wvdfile	"	Wave height time series file [Nx2] date/time in 'yyyymmddHHMM' and Hs, Tp, dir
swl0	0	Fixed still water level relative to MSL
Sediment limitations		
sedlim	1	switch for sediment limiter (0/1)
ldbsedlim	"	file with sediment limitation coordinates and properties, with [Nx3] specification.
xsedlim	[]	alternative input with x-coordinates of areas with sediment limitation separately [Nx1]
ysedlim	[]	alternative input with y-coordinates of areas with sediment limitation separately [Nx1]
widthsedlim	[]	alternative input with critical width for areas with sediment limitation [Nx1]
mud properties		
mud	0	mud transport option (0/1)
mudtaucr	0.3	critical shear stress for erosion (N/m2)
mudm	1.00E-04	erosion rate (kg/m2/s)
mudb	1000	muddy transport zone width (m)
mudmhw	1	mhw level [m]
mudmsl	0	msl level [m]
mudtfm	10	tfm value
mudw	1.00E-04	fall velocity of the muddy sediment [m/s]
mudbfcrit	500	critical mudflat width [m]
mudbmmin	1	minimum for the width of the muddy transport zone [m]

mudbmmax	100000	maximum for the width of the muddy transport zone [m]
ldbriverdisch	"	river discharge file with .riv extension, containing [xriv1, yriv1, xriv2, yriv2, tstart, tend, rate]
ldbmangrove	"	mangrove definition file with .mgv extension, containing [xmgv, ymgv, Bf, Bm, Bfm]
physics of spit width		
spitmethod	'default'	Overwash method
spitwidth	50	width of tip of spit (used for overwash)
spitheadwidth	200	width of tip of spit (used for upwind correction)
owscale	0.1	scales rate of the overwash per timestep (part of the deficit is moved to the backbarrier)
owtimescale	0	timescale for overwash (i.e. what part of the deficit is moved to the backbarrier)
spitdsf	d*0.8	underwater part of active height for shoreface -> used only in spit-width function
spitdbb	0.5*spitdsf	underwater part of active height for back-barrier -> used only in spit-width function
bheight	2	berm height used for overwash function (i.e. added to Dsf or Dbb)
tideinteraction	FALSE	tide interaction switch (0/1)
waveinteraction	FALSE	wave interaction switch (0/1)
wavefile	"	wave table (.mat)
surfwidth	1000	width of surf zone, where to update the bathymetry
bathyupdate	"	the dates when the bathymetry should be updated {'yyyy-mm-dd'}
channel		
channel	0	switch for migrating inlet (0/1)
channelwidth	550	target channel width
channelfac	0.08	adaptation factor, scales the response of the channel
channeldischrte	0	discharge rate
channeldischr	300	discharge rate of river
ldbchannel	[]	file with initial channel axis x and y coordinates, or directly a [Nx2] matrix (option 1)
rero	300	radius of flood delta influence on coast
rdepo	600	radius of inlet influence on flood delta
tscale	1	timescale of flood delta filling
xrmc	"	x-coordinates of rivers (option 2)
yrmc	"	y-coordinates of rivers (option 2)
flood delta		
flooddelta	0	switch (0/1) for flood delta
ldbflood	[]	wide outline of potential flood delta deposits
xfloodpol	[]	x-coordinates of flood delta polygon
yfloodpol	[]	y-coordinates of flood delta polygon
ldbspit	"	file with x and y coordinates of the spit polygon [Nx2] or directly provide a matrix
xspitpol	[]	x-coordinates of the spit polygon
yspitpol	[]	y-coordinates of the spit polygon
dx	50	resolution of flood delta area [m]
overdepth	2	initial overdepth flood delta [m]
formatting / output		
plotvisible	1	plot and update figure with wave conditions and modeled shoreline during run
xlimits	[]	x-limits of plot [m] [1x2]
ylimits	[]	y-limits of plot [m] [1x2]
xywave	[]	X,Y location and scale of wave arrow [m] [1x2]
xyoffset	[0,0]	shift in X,Y locaton for plotting

pauselength	[]	pause between subsequent timesteps (e.g. 0.0001)
outputdir	'Output\'	output directory
rundir	'Delft3D\def_model\'	run directory of a coupled Delft3D flow model
ldbplot	{}	cell array with 'LDB-filename', 'legend entry' and 'plot format' (e.g. 'b--')
plots	0	plot wave height at depth-of-closure (TDP) as coloured markers at every 'nth' grid cell
plotdir	0	plot wave direction at depth-of-closure (TDP) as quivers at every 'nth' grid cell
plots	0	plot transport rates at depth-of-closure (TDP) as colour markers/text at every 'nth' grid cell
plotupw	0	plot locations with high angle correction (use 0/1 as switch)
llocation	'SouthWest'	location of legend. Shortened for Octave compatibility
ld	3000	Width of the land fill behind the shoreline [m]
usefill	1	option switch that can be used to only plot lines instead of the fill
usefillpoints	0	option that can be used to force the model to use a specified number of landpoints
fignyear	12	the number of plots that need to be stored to a file each year [1/year]
plotinterval	1	the interval at which the coastline is plotted (in number of timesteps inbetween)
fastplot	1	use imwrite plotter, that is ~2x as fast as print
xyout	[]	specify output grids. Option 1: [x1,y1,x2,y2]. Option 2: a cell {} with [Nx2] xy-coordinates
xyprofiles	[]	specify multiple output profile locations [Nx2 with x,y; ...]
storageinterval	50	time interval of storage of output data to a file ('output.mat')
extract shorelines		
slplot	{}	slplot
extractxy	0	get file with shorelines coordinates x,y
printfig	0	switch to print figures to files (0/1)
extract shoreline & dune foot locations		
yesplot	0	only needed for interactive mode, not for regular runs with plots
bermwplot	0	for extracting against certain times for a certain transect(s).
bermwplotint	[]	beach berm width plot interval [Months]
qplot	0	to plot wave and wind transport at each time step for a certain transect(s).
transect	"	file indicies x-y transects to be plotted
clplot	0	switch to track coastline location relative to the initial coastline at transects
clplotint	[]	coastline change plot interval a certain transect(s) [Months] .
extractbermplot	0	to allow data extraction for berm width plotting
video		
video	0	switch for capturing a video (0/1)
debug		
debug	0	switch for setting the debug modus (0=off, 1-5 different storage / plot options)
data Assimilation		
da	0	data assimilation switch (0/1)
bs	0	switch of brier-skill score (0/1)